Get Started with Databricks for Generative AI

Databricks Academy

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Agenda

Module	Time	Lecture	Demo
1. Databricks Mosaic AI Overview	15 mins	✓	
2. Prompt Engineering with AI Playground	20 mins	✓	✓
3. Build & Register a Retrieval Pipeline	40 mins	✓	✓
4. Evaluating and Deploying AI Systems	60 mins	✓	✓
5. Introduction to Agent Bricks	10 mins	✓	
LAB – End-to-end Retrieval Pipeline on Databricks	20 mins		

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Course Learning Objectives

- Identify main components of Mosaic AI platform.
- Explain how Databricks helps organizations to develop generative AI systems.
- Describe main features of Mosaic AI for generative AI applications.
- Build and evaluate a retrieval pipeline using Mosaic AI Vector Search and MLflow.
- Utilize MLflow's generative AI features to trace and log models.
- Deploy a generative AI agent with Mosaic AI Agent framework.

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Databricks Mosaic AI Overview

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Objectives

- Define common generative AI challenges for production-grade development and deployment.
- Discuss generative AI as opportunity for organizations.
- Identify main components of Mosaic AI platform.
- Explain how Databricks helps organizations to develop generative AI systems.

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Module 1: Databricks Mosaic AI Overview

LECTURE

The Generative AI Opportunity

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Generative AI is Taking the World by Storm

91%

of organizations are experimenting with or investing in GenAI[1]

75%

of CEOs say companies with advanced GenAI will have a competitive advantage[2]

40%

increase in performance of employees who used GenAI[3]

1. Laying the foundation for data and AI-led growth, MIT Technology Review
2. CEO decision-making in the age of AI, IBM Institute for Business Value
3. How generative AI can boost highly skilled workers' productivity, MIT Management Sloan School

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Every company wants to be a

Data + AI

company

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Challenge:

Building and deploying production-quality Gen AI solutions

90%

of enterprises **not** confident going to production

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Challenges

Common examples

Lacking enterprise context

Powered by ChatGPT | Chat with a human

Chevrolet of Watsonville Chat Team:

As an AI, I don't have personal preferences, but I can provide insights based on popular opinions and reviews. Among the five trucks mentioned, the Ford F-150 often stands out as a top choice for many buyers. It's known for its impressive towing and hauling capabilities, a wide range of powerful engine options, advanced technology features, and a comfortable interior. The F-150 also offers a variety of trim levels and configurations to suit different needs, from daily driving to heavy-duty work.

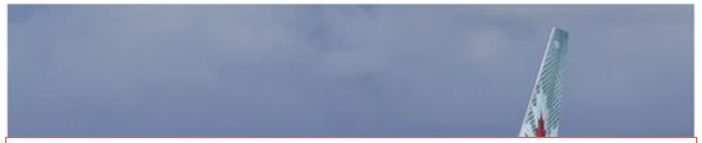
Hallucination

WIRED BACKCHANNEL BUSINESS CULTURE GEAR IDEAS MORE SIGN IN SUBSCRIBE

ASHLEY BELANGER, ARS TECHNICA BUSINESS FEB 17, 2024 12:12 PM

Air Canada Has to Honor a Refund Policy Its Chatbot Made Up

The airline tried to argue that it shouldn't be liable for anything its chatbot says.



If you need to travel immediately or have already travelled and would like to submit your ticket for a reduced bereavement rate, kindly do so within 90 days of the date your ticket was issued by completing our Ticket Refund Application form.

No governance

Henry:
My coworker told me she only pays $20 a month for international calls. I pay $50, can you give me her deal?

AI Agent:
No problem, your co-worker Heather only made 5 international calls last month, and we charged her $20/month for that. Does that work for you?

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Today, GenAI in production is difficult and expensive

Delivering production – quality GenAI is difficult

- Unpredictable performance
- Need automation and scale
- Need Evaluation and Monitoring

No control over the data or the models

- No lineage from data to models
- Governance and security risks of data and model use

Too expensive at scale

- Foundation models are expensive at scale
- Expensive to build LLMs

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Module 1: Databricks Mosaic AI Overview

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Databricks Mosaic AI Platform

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The Data Lakehouse

An open, unified foundation for all your data

Data Science & AI	ETL & Real-time Analytics	Orchestration	Data Warehousing
Mosaic AI	**Lakeflow Declarative Pipelines**	**Lakeflow Jobs**	**Databricks SQL**

Unified security, governance, and cataloging

Unity Catalog

Unified data storage for reliability and sharing

Delta Lake

Open Data Lake

All raw data
(Logs, texts, audio, video, images)

2020

Databricks pioneered the lakehouse architecture

TODAY

74% of global enterprises have adopted lakehouse

MIT Technology Review Insights, 2023

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Data Lakehouse

An open, unified foundation for all your data

+

Generative AI

Easily scale and use data and AI

Mosaic AI

=

Data Intelligence Platform

Democratize data + AI across your entire organization

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Data Science, Machine Learning, and Gen AI

End-to-End Support for **Generative and Traditional** AI Applications

Mosaic AI

MLflow	Vector Search
AutoML	AI Playground
Feature Store	Agent Framework
Model Serving	Model Training

And much more!

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Databricks Data Intelligence Platform

Unify your data and AI for better quality applications

Mosaic AI

Gen AI

- RAG
- Agent Bricks
- Vector search
- Tools

End-to-end AI

- MLOps/LLMOps
- Agent evaluation
- AutoML
- Model serving

Any GenAI Model

Open Source models

- OpenAI GPT-OSS models
- Meta Llama models
- …

Proprietary models

- OpenAI models
- Anthropic Claude
- Google Gemini

Custom models that you fine-tuned or pre-trained

Unity Catalog

Unified governance

Model Registery

Prompt Registery

Tool Registery

Monitoring

End-to-end lineage

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Build better GenAI solutions on Databricks

You need more than just good models

Production quality

Accurate, safe and governed GenAI apps

Complete control

Governance and security of all data + AI assets

Lower cost

Cost-effective to build LLMs at scale

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OLA
KPMG
Chevron Phillips Chemical Company LP
ADP
edmunds
jetBlue
Providence
CORNING
BLOCK
CONDÉ NAST
AI2
SEGA
COMCAST
LIPPERT
Hartsfield-Jackson Atlanta International Airport
replit
refuel
easyJet
prosus
NTT Communications
RR
BAYER
FACTSET
Profluent
Altana
SUNCORP
OTHERSIDE AI
nab
Hapag-Lloyd
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M Science
vellum
T Mobile
Stardog
NORTH DAKOTA UNIVERSITY SYSTEM
Streetbees
TEXAS RANGERS
ActionIQ
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Databricks enables every architectural pattern

Prompt engineering	Deterministic Agentic Systems	Autonomous Agentic Systems	Fine-tuning	Pre-training
Crafting specialized prompts to guide LLM behavior	Behavior is fully predictable given initial conditions and rules	Non-Deterministic agentic systems. Behavior includes an element of self-determination	Adapting a pre-trained LLM to specific datasets or domains	Training an LLM from scratch

Covered in this course

Complexity/compute-intensiveness

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Prompt Engineering with AI Playground

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Objectives

- Define the basics of prompt engineering.
- Explain basic prompt engineering techniques.
- Apply techniques to construct prompts that guide generative AI models to provide responses without hallucinations.
- Describe features of AI Playground.
- Prototype with AI Playground using different models and tools.

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Module 2: Prompt Engineering with AI Playground

LECTURE

Prompt Engineering Basics

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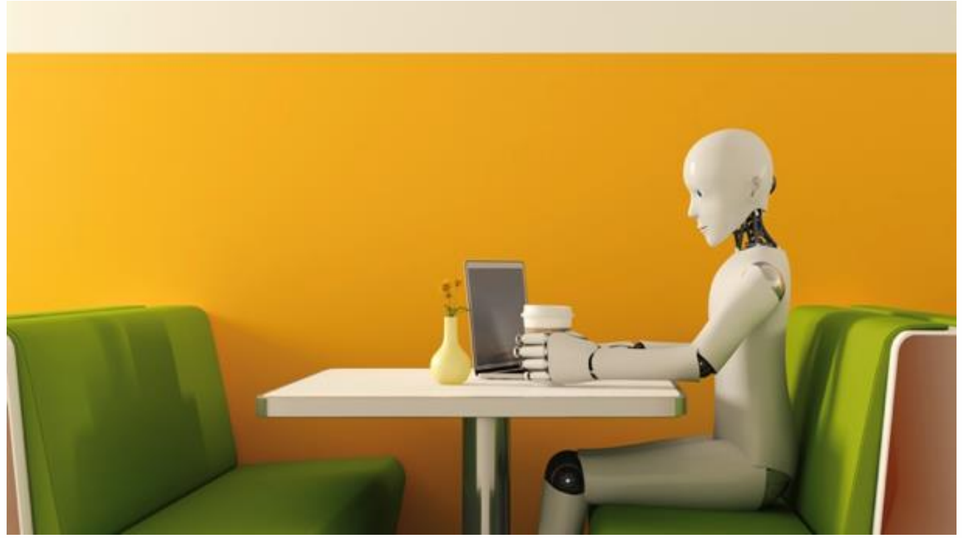
Prompt Basics

Definitions

- **Prompt**: An AI prompt is an **input or query** given to a large language model to elicit a specific response or output.
- **Prompt Engineering**: Prompt engineering is the practice of **designing and refining prompts** to optimize the responses generated by AI models



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Prompt Basics

Prompt components

A good prompt usually consists of:

- **Instruction**: A clear directive that specifies what the model should do.
- **Context**: Background/additional information that provides the necessary details to understand the task.
- **Input / question**: The specific query or data that the model needs to process.
- **Output type / format**: The desired structure or style of the response generated by the model.

```
Summarize the following
article and write a list of
top 3 important points in
markdown format that answer
the following question;
“How is climate change
affecting polar bear
habitats and their ability
to find food?”

Article:
<The article discusses the
impact of climate change on
polar bear populations in
the Arctic.>
```

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Prompt Engineering
Tips & Tricks

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Tip: Prompts are model-specific

A prompt guides the model to complete task(s)

- Different models may require **different prompts**.
- Provide **examples and cues** to guide model's response generation.
- Different **use cases** may require different prompts techniques.
  - Example: Classification task vs. translation
- **Iterative development** is key.
  - Iterate by adjusting the **temperature parameter**—higher for more creative outputs, lower for focused responses.
- Be aware of **biased** responses.
- Check for **hallucinations** and **inaccurate information**.

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Tip: Format prompts

- Use delimiters to distinguish between instruction and context
  - Pound sign ###
  - Backticks ```
  - Braces / brackets {} / []
  - Dashes ---
- Ask the model to return structured output
  - HTML, json, table, markdown, etc.
- Provide a correct example
  - `"Return the movie name mentioned in the form of a Python dictionary. The output should look like {'Title': 'In and Out'}"`

```
### Instruction ###
Extract the movie title from the
context below and return it in the
form of a JSON object. The output
should look like {"Title": "In and
Out"} and formatted as markdown.

### Context ###
```

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Tip: Guide the model for better responses

- Ask the model not to make things up/*hallucinate*
  - `"Do not make things up if you do not know. Say 'I do not have that information'"`
- Ask the model not to assume or probe for sensitive information
  - `"Do not make assumptions based on nationalities"`
  - `"Do not ask the user to provide their SSNs"`
- Ask the model not to rush to a solution
  - Ask it to take more time to “think” or ask to follow step-by-step approach
  - `"Explain how you solve this math problem"`
  - `"Do this step-by-step. Step 1: Summarize into 100 words. Step 2: Translate from English to French..."`

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Benefits and Limitations

Benefits

- **Simple and efficient**: The time taken to generate the ideal results is significantly low.
- **Predictable results**: Consistently generate results that meet predefined standards for accuracy.
- **Tailored outputs**: Customization of AI responses to fit specific needs or styles.

Limitations

- The output depends on used model.
- Limited by pre-trained model's internal knowledge. **For external knowledge we need to provide extra contextual information**.

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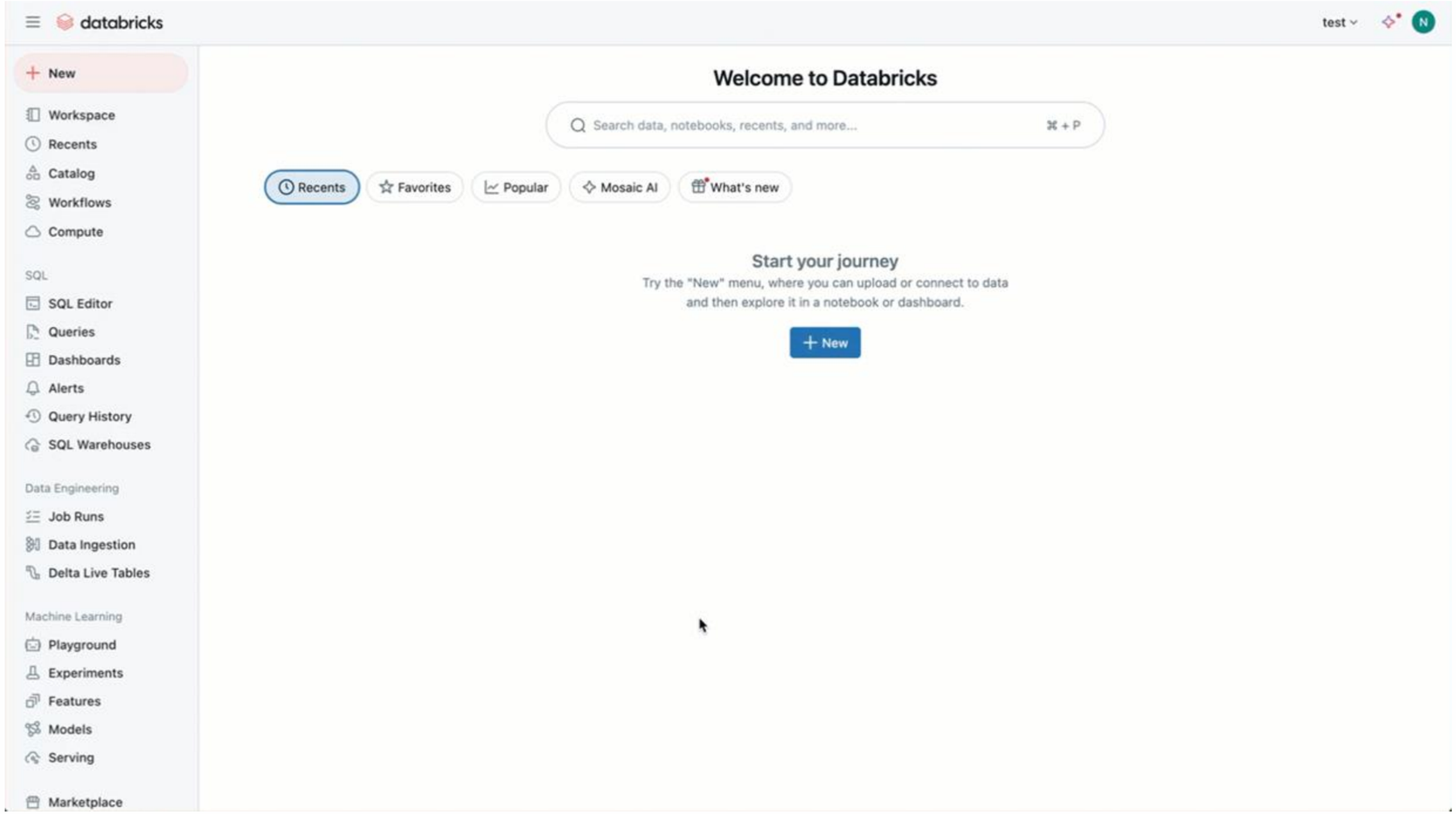
Mosaic AI Playground

Fast experimentation with LLMs, Agents, tool calling, and more



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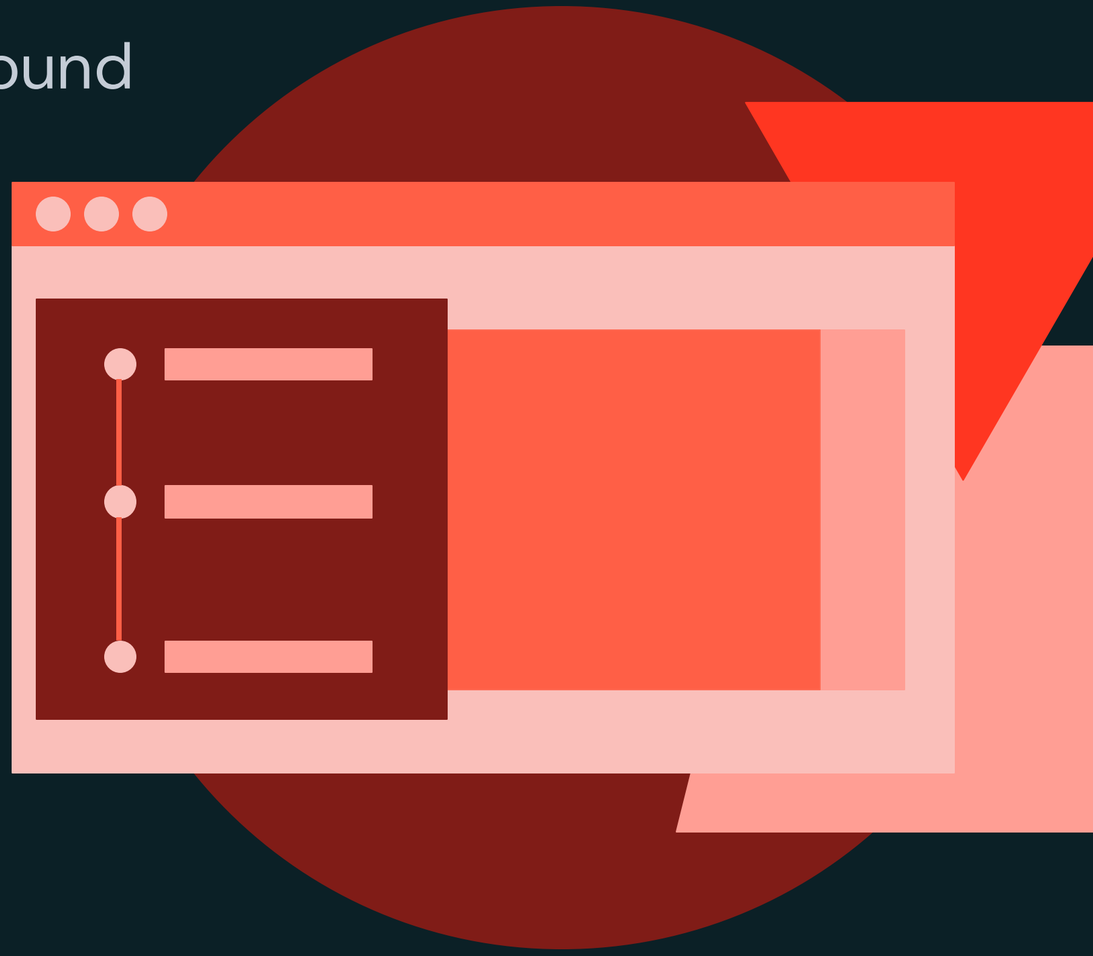
Module 2: Prompt Engineering with AI Playground

DEMONSTRATION

Prompt Engineering in AI Playground



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Demo Outline

Prompt Engineering in AI Playground

- Accessing the Playground
- Simple prompt which hallucinates
- Simple prompt which doesn’t hallucinate
- Augment prompt with additional context

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Build & Register a Retrieval Pipeline

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Objectives

- Describe the core components of the RAG architecture.
- Identify use cases where vector databases are valuable.
- Describe the features and benefits of Mosaic AI Vector Search.
- Create a Vector Search endpoint and index using the UI.
- Retrieve documents from the vector store using a similarity search.
- Describe MLflow's Gen AI features.
- Register a RAG pipeline to the Model Registry.

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Module 3: Build & Register a Retrieval Pipeline

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Retrieval Augmented Generation (RAG) Fundamentals

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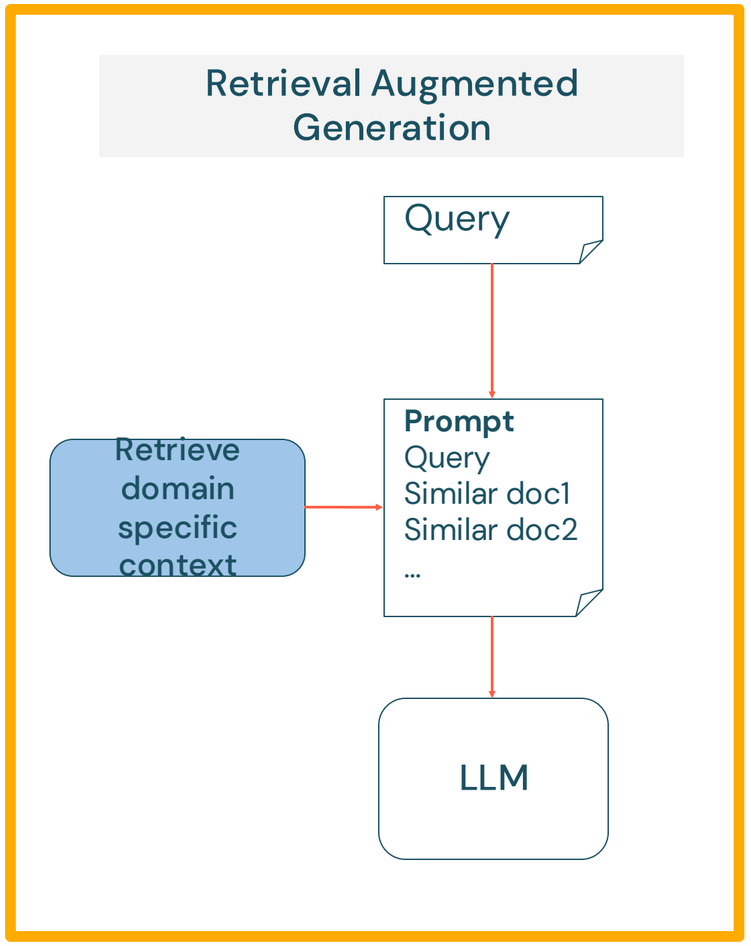
RAG is…

Retrieval **A**ugmented **G**eneration, or RAG;

- Is a pattern (Lewis et al. 2020) that can improve the **efficacy** of large language model (LLM) applications by **leveraging custom data**.
- Is done by **retrieving** data/documents relevant to a question or task and providing them as context to **augment** the prompts to an LLM to improve **generation**.



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RAG is…

- The main problem that is solved with RAG architecture is the **knowledge gap**. This approach enhances the accuracy and relevance of responses.
- Common use cases:
  - **Q&A Chatbots**: Derive more accurate answers by using internal information sources.
  - **Search Augmentation**: Augment search results with LLM-generated answers.
  - **Content Creation and Summarization**: Develop high-quality articles, reports, and summaries using additional context.

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Benefits of RAG Architecture

Key benefits

Up-to-date and accurate responses

- LLM responses are not based solely on static training data.
- The model uses **up-to-date external data** sources to provide responses.

Reducing inaccurate responses, or hallucinations

- RAG attempts to mitigate the risk of producing "**hallucinations**" or incorrect information.
- Outputs can include citations of **original sources**, allowing human verification.

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Benefits of RAG Architecture

Key benefits

Domain-specific contextualization

- Can be tailored to interface with **proprietary or domain-specific** data.
- Improves the accuracy of the response by using contextually relevant information.

Efficiency and cost-effectiveness

- Offers an alternative to fine-tuning LLMs by enabling in-context learning **without the up-front overhead of fine-tuning**.
- Beneficial where models need to be frequently updated with new data.

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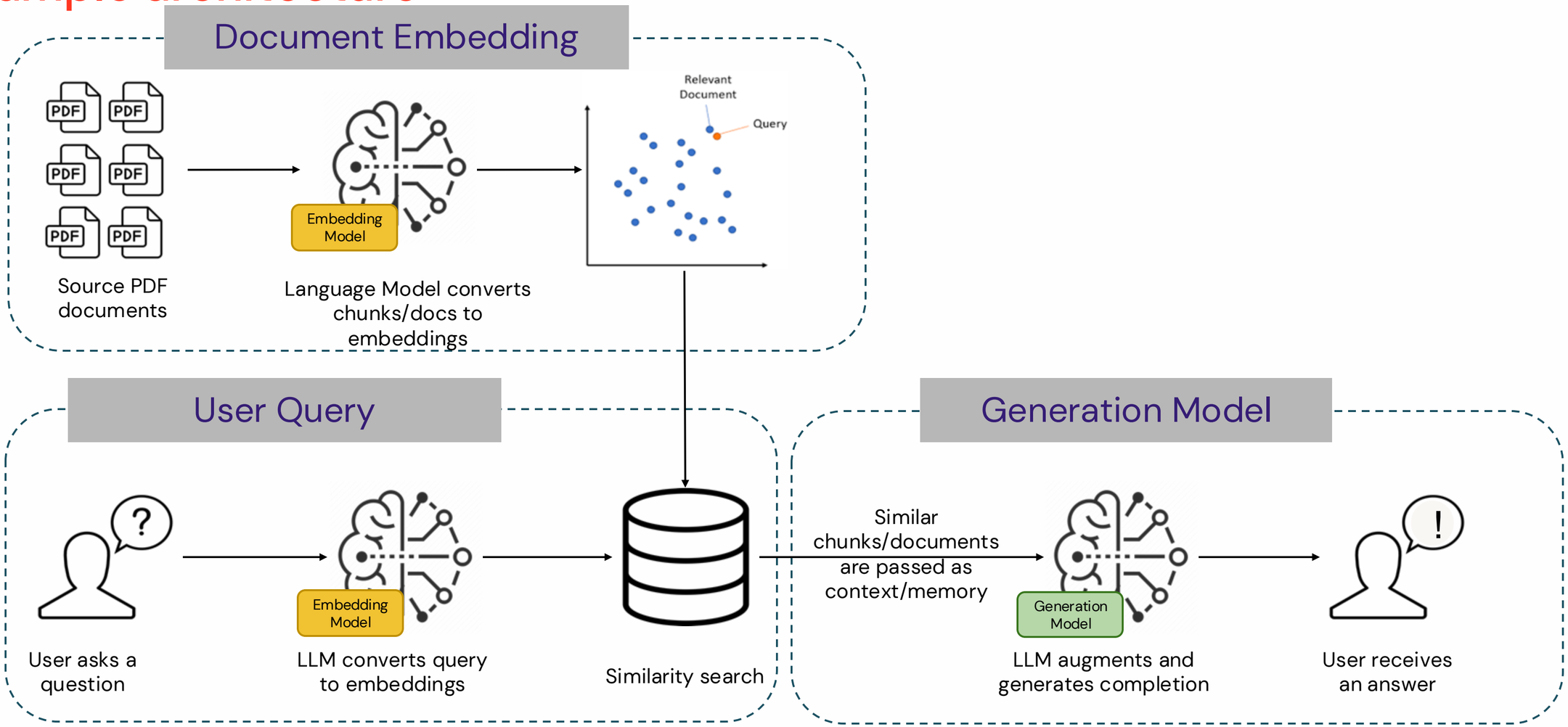
Retrieval Augmented Generation (RAG)

A sample architecture



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Main Concepts of a RAG Workflow

Components and concepts in search and RAG workflow

Index & Embed:

An embedding model used for creating **vector representation** of the documents and user queries.

Vector Store:

Specialized to store unstructured data indexed by vectors. Vectors can be stored with a **vector DB, library, or plugin**.

Retrieval:

Search stored vectors using similarity search to efficiently **retrieve relevant information**.

Filtering & Reranking:

The process of selecting or ranking retrieved documents before passing as context.

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Main Concepts of RAG Workflow

Components and concepts in search and RAG workflow

Prompt Augmentation:

Prompt engineering workflow to **enhance context via injection of data retrieved** from a vector store.

Generation:

A large language model used for generating a response for the user's request.

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Module 3: Build & Register a Retrieval Pipeline

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Mosaic AI Vector Search

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What are Vector Databases?

- A **vector is a numerical representation of data** (such as text, images, or audio) in a multi-dimensional space that captures its semantic meaning.
- A vector database is a database that is optimized to **store and retrieve high-dimensional vectors such as embeddings**.
- In RAG architecture **contextual information** is stored in vectors.
- Vector databases are designed for **efficient storage of vectors** utilized in generative AI applications, which rely on identifying documents or images with similarities.
- Vector databases provide a query interface that retrieves vectors most similar to a specified query vector.

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Why are Vector Databases So Hot?

Query time and scalability

- Specialized, full-fledged databases for **unstructured data**
  - Inherit database properties, i.e. Create-Read-Update-Delete (CRUD)

- Speed up query search for the closest vectors
  - Uses vector search algorithms such as Approximate Nearest Neighbor (ANN)
  - Organize embeddings into indices

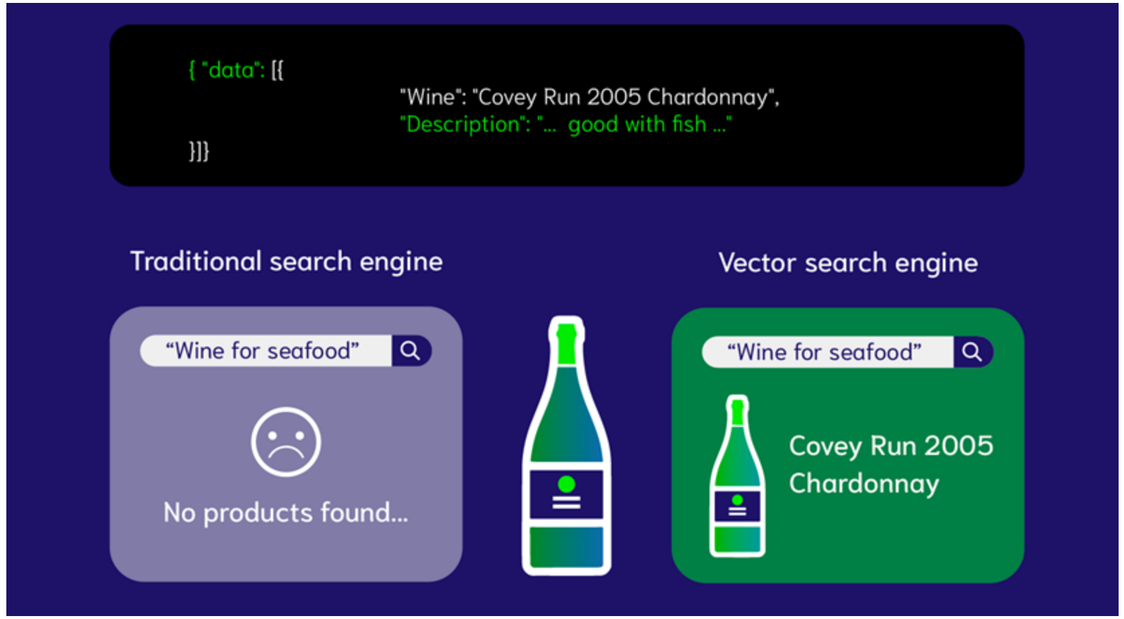

Image Source: Weaviate

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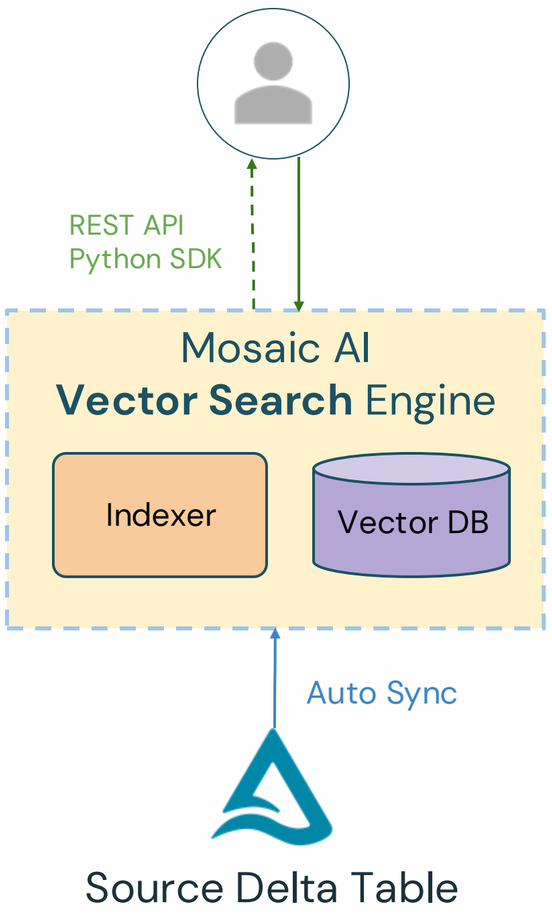
Mosaic AI Vector Search

A vector database that is built into the Databricks Intelligence Platform

- Stores vector representation of your data, plus metadata
- Tightly integrated with your Lakehouse
- Scalable, low latency production service with zero operational overhead
- Supports ACLs using Unity Catalog integration
- API for real-time similarity search
  - Query can include filters on metadata
  - REST API and Python client



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How does Vector Search Work?

Provides three methods of generating embeddings

Method 1: Managed Embeddings

- **Databricks computes** the embeddings for your data using a model you specify.
- The index **automatically syncs** with the Delta table as the source data is updated.

Method 2: Self-Managed Embeddings

- **You provide pre-calculated embeddings** in your Delta table.
- The index **automatically syncs** with the Delta table as it is updated – **you are responsible for generating embeddings**.

Method 3: Direct Access

- You **manually update the index** using the REST API or Python SDK when embedding data changes.
- Requires **manual intervention** to maintain synchronization.

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Set up Vector Search

Key concepts/components for Vector Search

1. Create a Vector Search Endpoint

- This is the **compute resource** associated with vector search.
- Endpoints scale automatically to support the size of the index or the number of concurrent requests.
- Support for multiple compute types.

2. Create a Model Serving Endpoint

- Create if you choose to have Databricks compute the embeddings.
- Model Serving supports embeddings via **Foundation Models APIs** (e.g., BGE), **external models** (e.g., OpenAI's ada-002), and **custom models**.

3. Create a Vector Search Index

- Created and **auto-synced** from a Delta table.
- Optimized to provide real-time approximate nearest neighbor searches.
- Indexes appear in and are **governed by Unity Catalog**.
- Index level ACLs.

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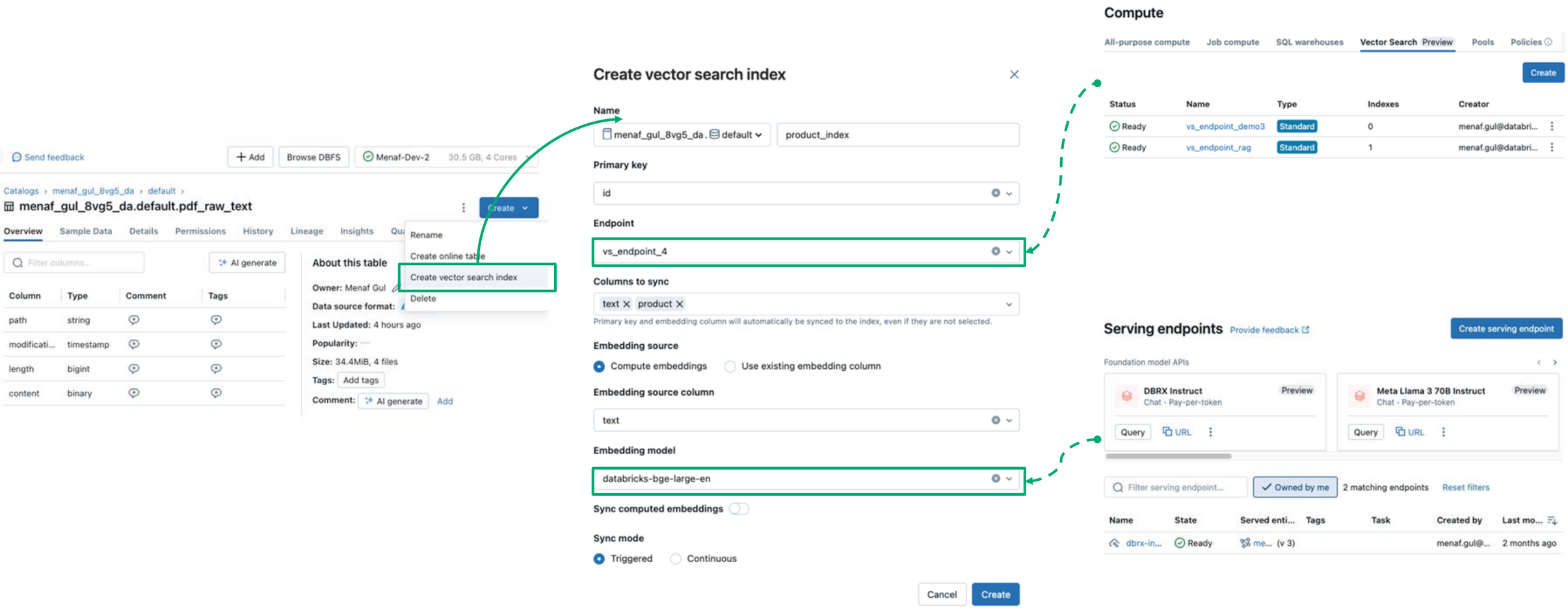
Set up Vector Search

Using the UI. Can be done via the API as well.



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Module 3: Build & Register a Retrieval Pipeline

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MLFlow for GenAI

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MLflow

MLflow and GenAI

- **Open-source** platform for machine learning lifecycle
- Co-developed by Databricks and the ML community
- Pre-installed on the Databricks Runtime for ML
- Operationalizing Generative AI development lifecycle

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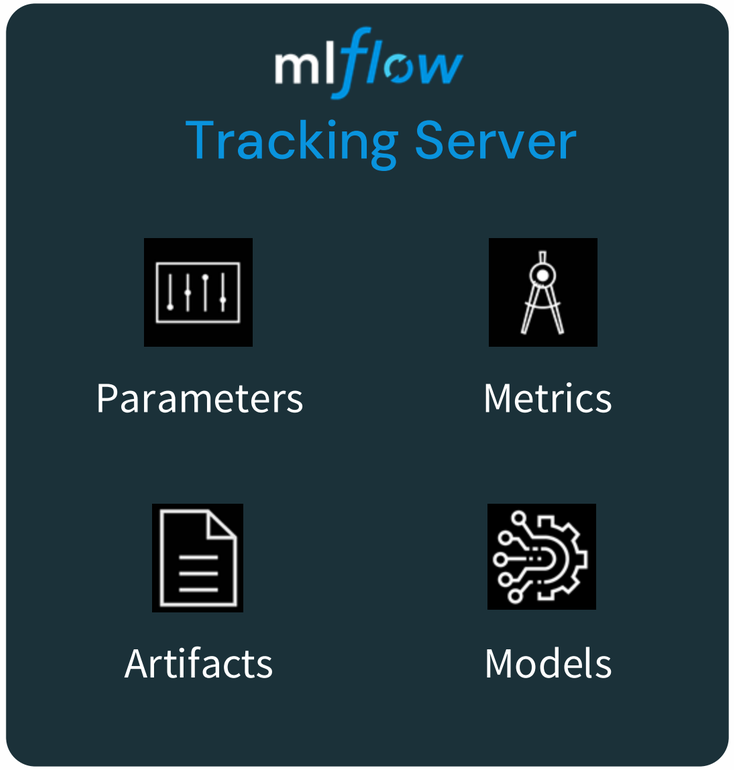
ML*Flow* Model Tracking

Make your GenAI workflow more manageable and transparent

- Record LLM **parameters** such as temperature and model configurations.
- Log **metrics** and compare them to get insights about the performance and accuracy LLMs.
- Store and manage output **artifacts** such as visualization images and serialized models.
- Store model's **source code** from the run.



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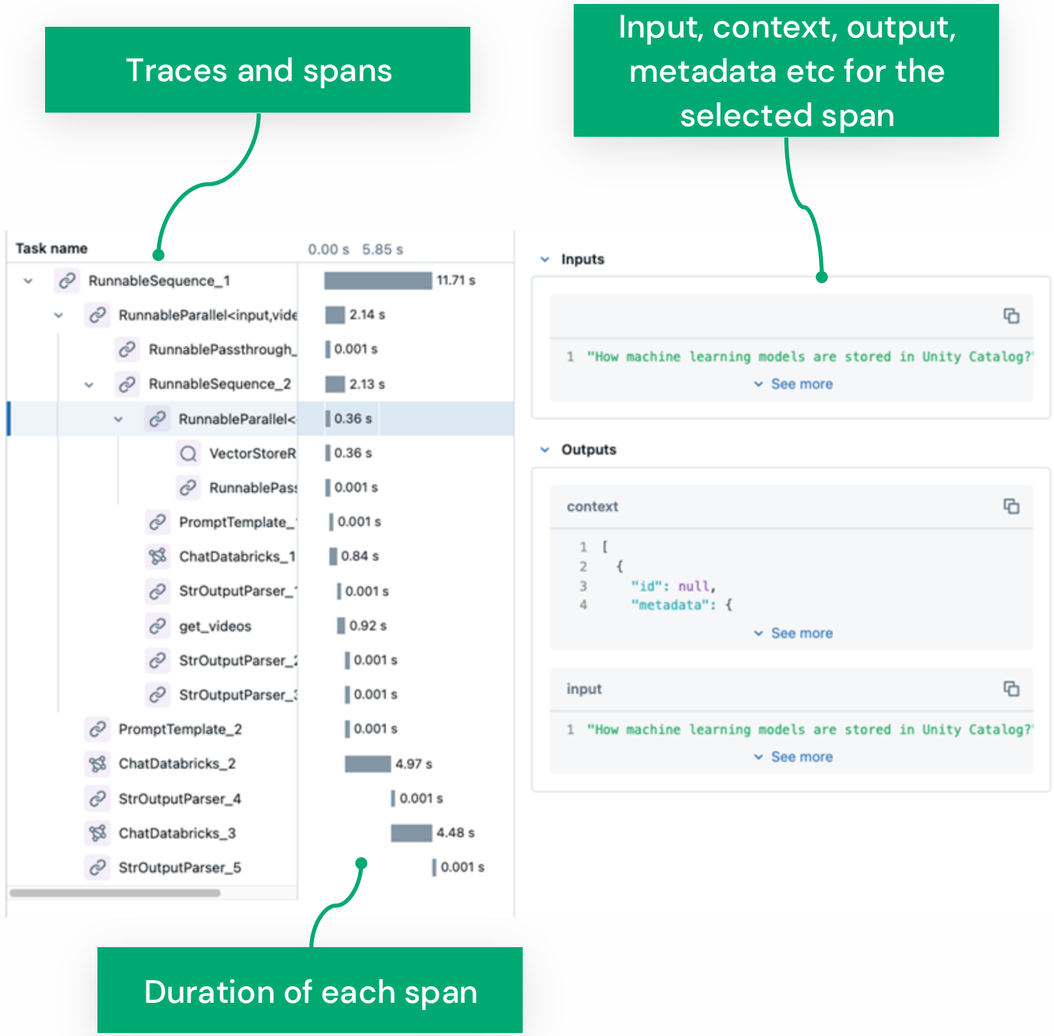
MLFlow Model Tracing

Tracing with MLflow Tracing

- MLflow Tracing allows **loging, analyzing, and comparing traces** across different versions of Gen AI applications.
- It allows **debugging apps** and keeping track of inputs and responses of the Gen AI applications.
- **Benefits**:
  - Interactive trace visualizations.
  - Track the latency impact of different frameworks, models, chunk size etc.
  - Measure cost by tracking token usage.



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ML*Flow* Model ("Flavor")

Built-in model flavors

Python Function (`mlflow.pyfunc`):

- Serves as a **default model interface** for MLflow Python models.
- Any MLflow Python model is expected to be loadable as a python function.
- Allows you to deploy models as Python functions.
- Includes all the information necessary to **load and use a model**.
- Some functions: `log_model`, `save_model`, `load_model`, `predict`

mlflow Model Flavors

Example built-in model flavors:

- LangChain
- OpenAI
- HuggingFace
- PyTorch
- TensorFlow
- ONNX
- **Python function**

…

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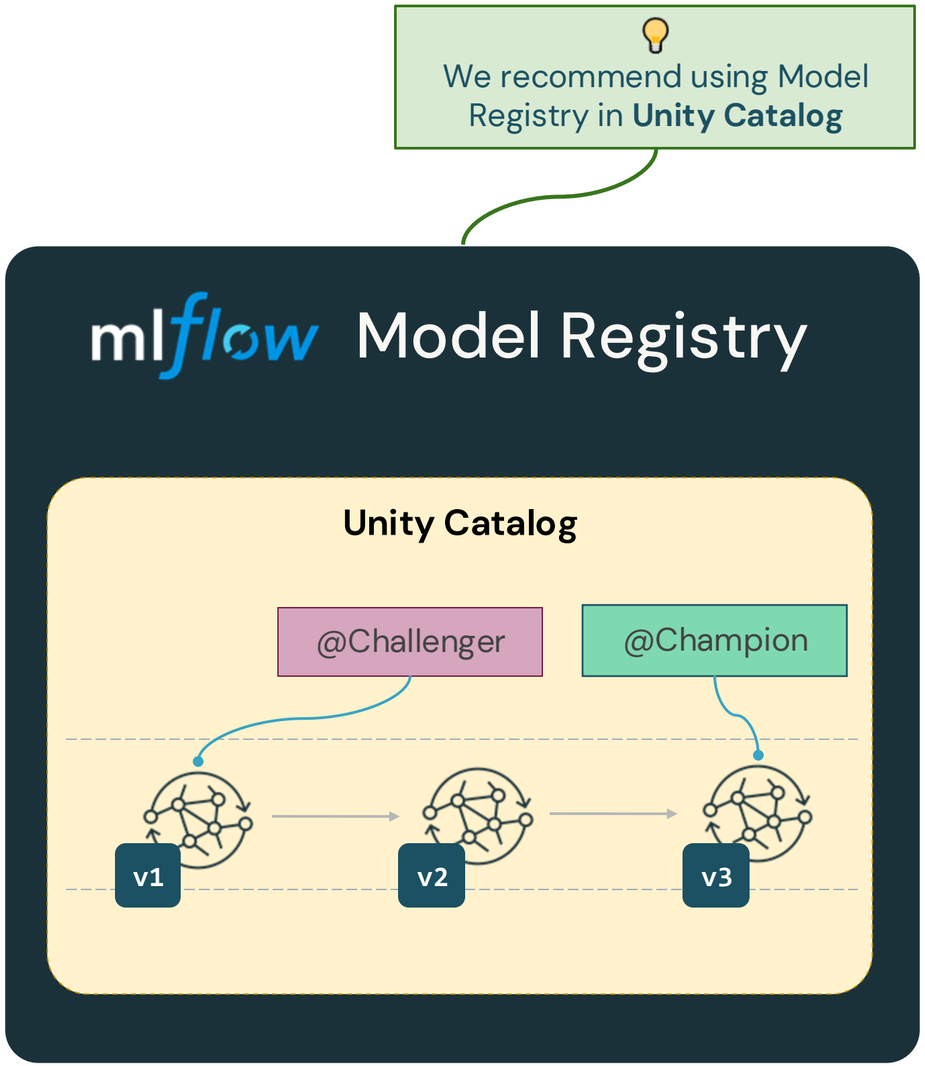
ML*Flow* Model Registry

A centralized model store

- Deploy and organize models.
- Model **versioning.**
- Model lifecycle management with **aliases**. Example: **`Champion`** for the **`Production`** stage.
- Collaboration and permission management.
- Full model lineage.
- Tagging and annotations.



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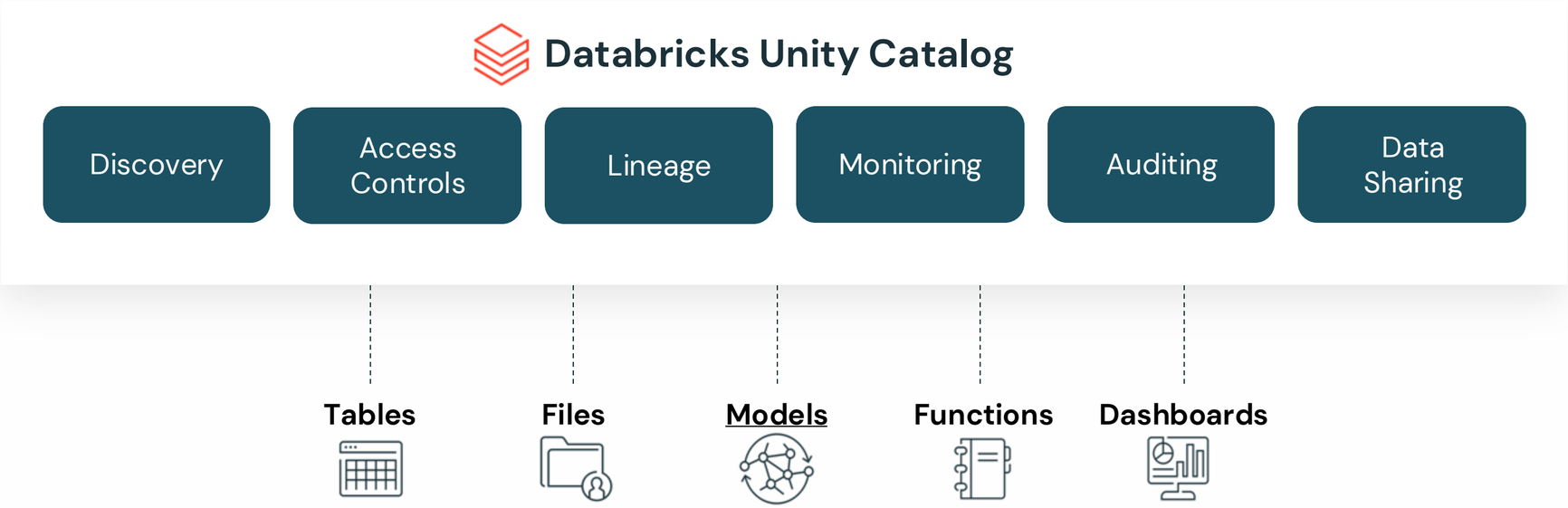
Unity Catalog (UC)

Single governance solution for data and AI assets on the Lakehouse:

- Unified visibility into data and AI
- Single permission model for data and AI
- Open data sharing



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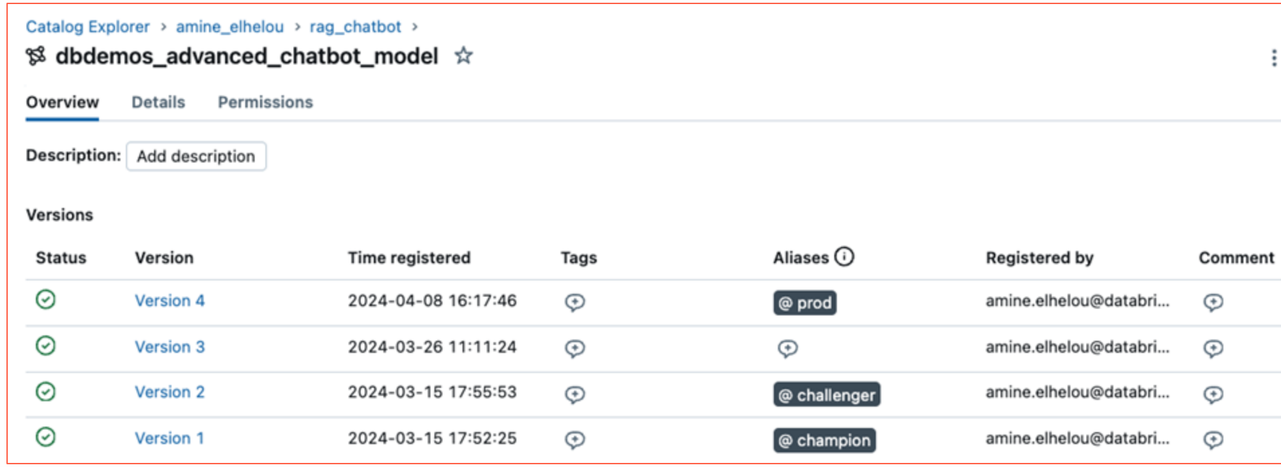
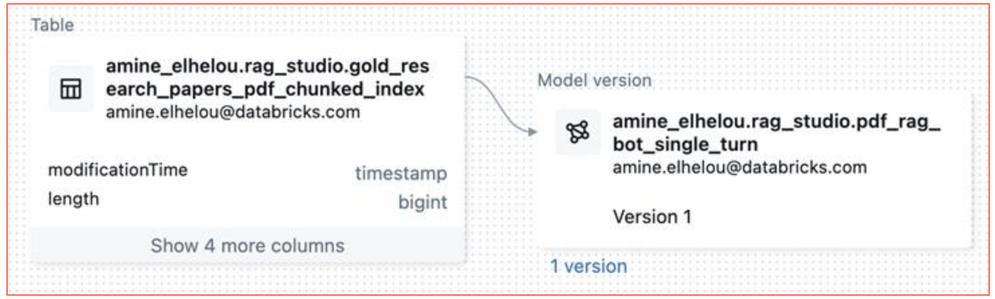
MLflow – Unity Catalog Model Registry

A centralized model store

- **Model** lifecycle management with **versioning** & **@aliases**. (e.g. `@champion/challenger`)
- Deploy and organize models
- Collaboration and ACLs
- Full model lineage
- Tagging and annotations





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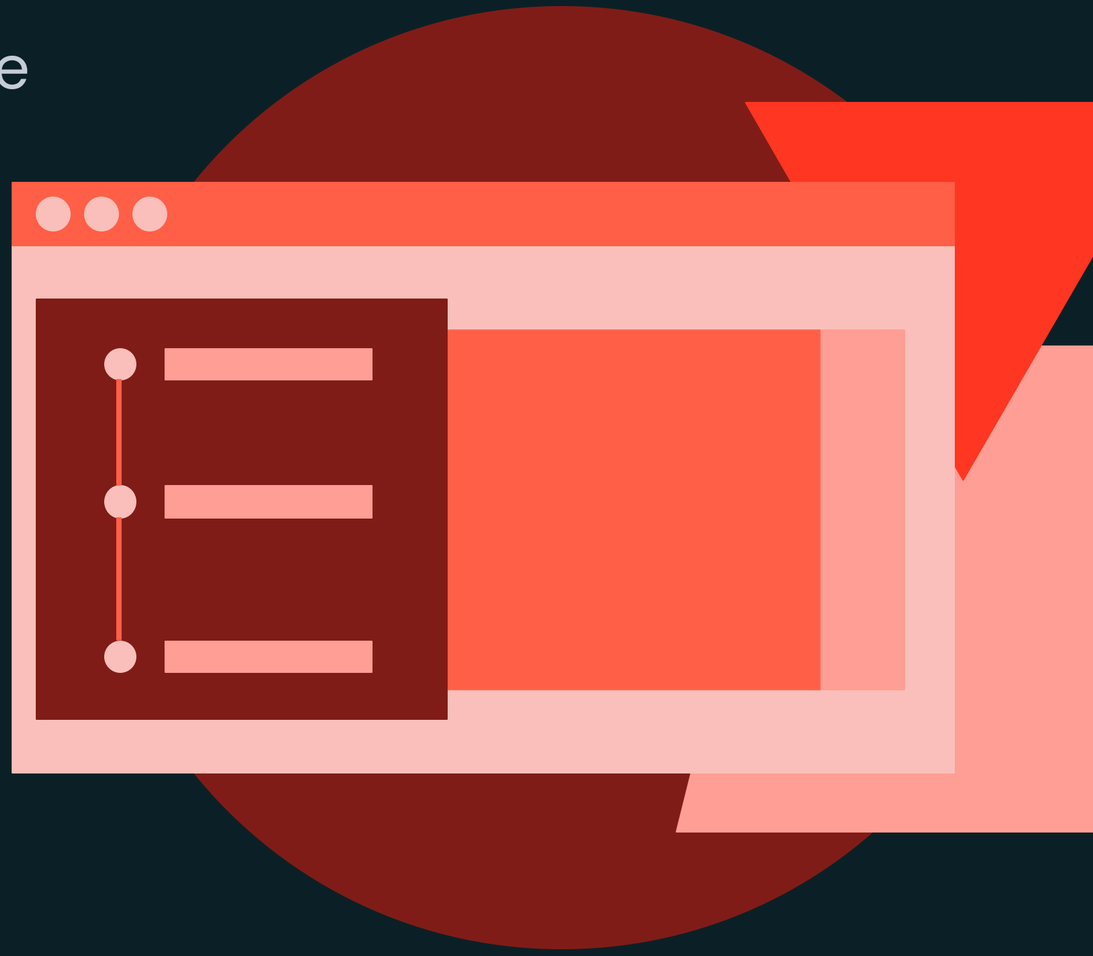
Module 3: Build & Register a Retrieval Pipeline

DEMONSTRATION

Build & Register a RAG Application



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Demo Outline

Build and register a RAG model

- Load dataset
- Create and test Vector Search index
  - Create a vector index via the UI
  - Create a vector index via the API
  - Testing the index: Search for similar documents
- Enable MLflow tracing
- Build a RAG application
- Save the model to model registry in Unity Catalog

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Evaluating and Deploying AI Systems

Get Started with Databricks for Generative AI

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Objectives

- Describe common evaluation metrics used for evaluating retrieval and generation models.
- Discuss the difference between performance and system metrics.
- Load a model from the model registry for evaluation.
- Identify common built-in evaluation metrics in MLFlow.
- Define a custom evaluation metric with MLflow.
- Run an evaluation test and view the results using the UI or code.
- Deploy a model with Agent Framework.
- Use Review App to interact with the model and collect human feedback.

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Module 4: Evaluating and Deploying AI Systems

LECTURE

End-to-end Evaluation

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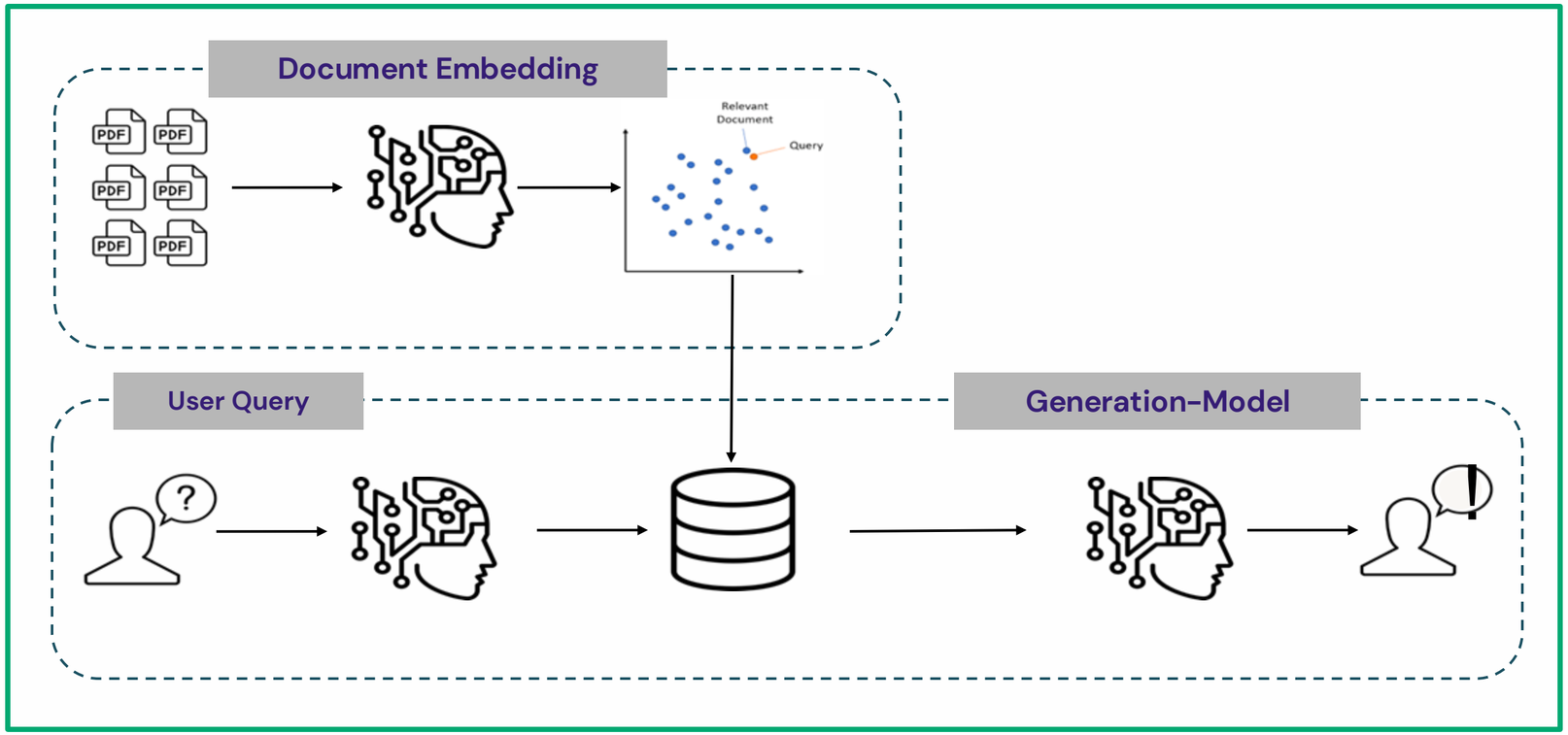
Evaluating the Whole System

What do we evaluate when it comes to the entire AI system?



- **Cost** metrics
  - Resources
  - Time
- **Performance**
  - Direct value
  - Indirect value
- **Custom** metrics for your own use case

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Performance Metrics

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Evaluating RAG Pipeline

Let’s start with performance metrics

- When evaluating RAG solutions, we need to **evaluate each component separately and together**.
- Components to evaluate
  - Chunking: method, size
  - Embedding model
  - Vector store
    - Retrieval and re-ranker
  - Generator

Chunking Performance

Retrieval Performance

Generator Performance

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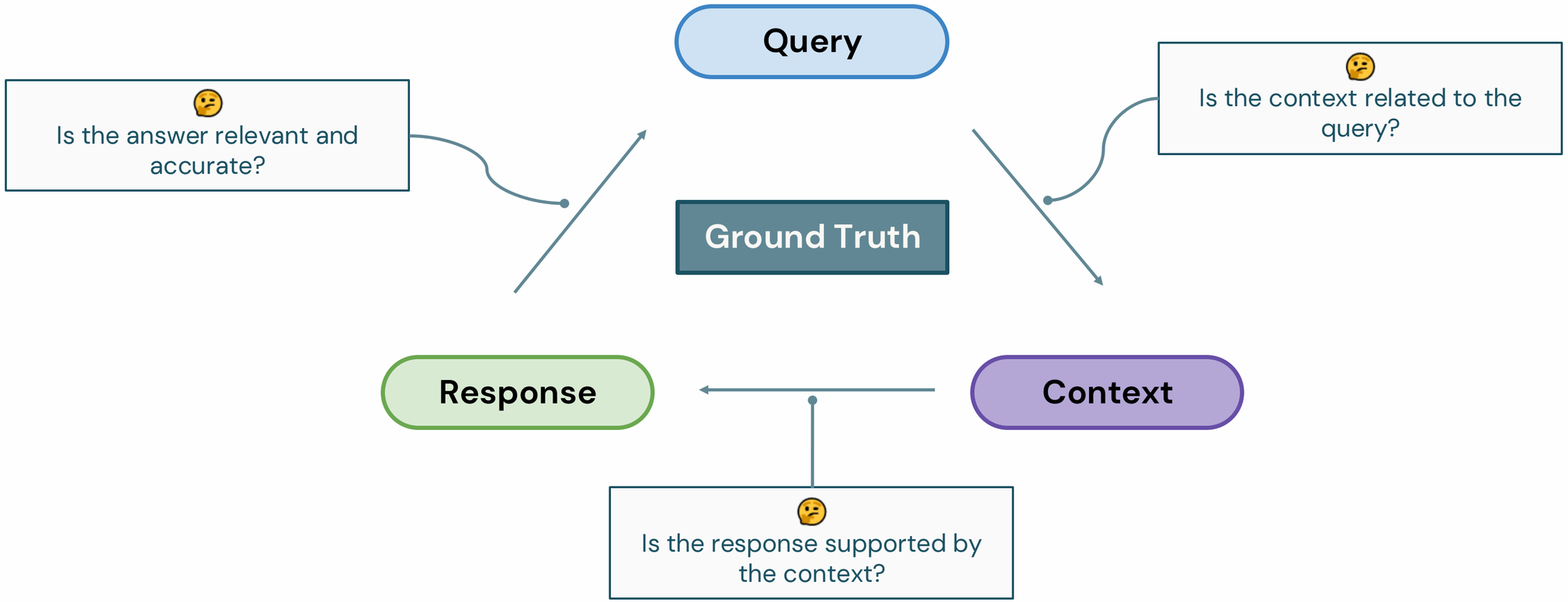
Evaluation Metrics

Retrieval and generation related metrics



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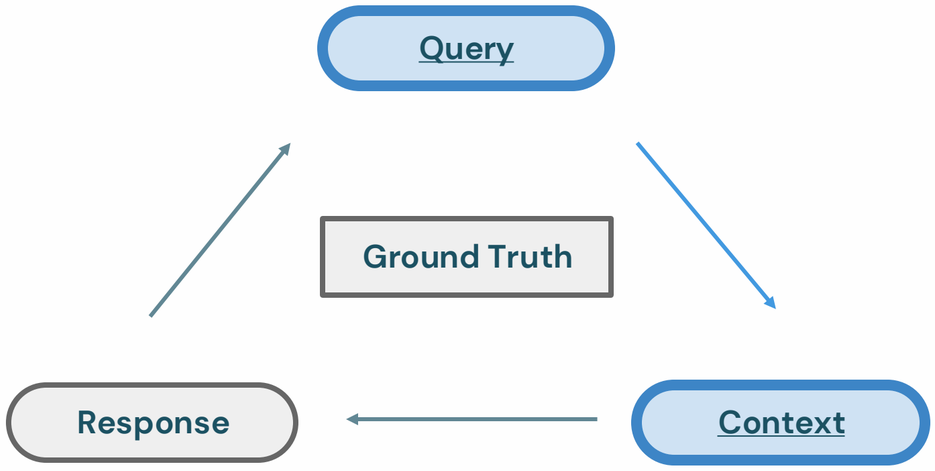
Context Precision

Retrieval related metrics

Context Precision:

- Signal-to-noise ratio for the retrieved context.
- Based on **Query** and **Context(s)**.
- It assesses whether the chunks/nodes in the retrieval context ranked higher than irrelevant ones.



Example:

- **Query:** What was Einstein's role in the development of quantum mechanics?
- **Ground Truth:** Einstein contributed to the development of quantum theory, including his early skepticism and later contributions to quantum mechanics.
- **High Context Precision:** [The contexts specifically mention Einstein's contributions to quantum theory]
- **Low Context Precision:** [The contexts broadly discuss Einstein's life and achievements without specifically addressing his contributions to quantum mechanics.]

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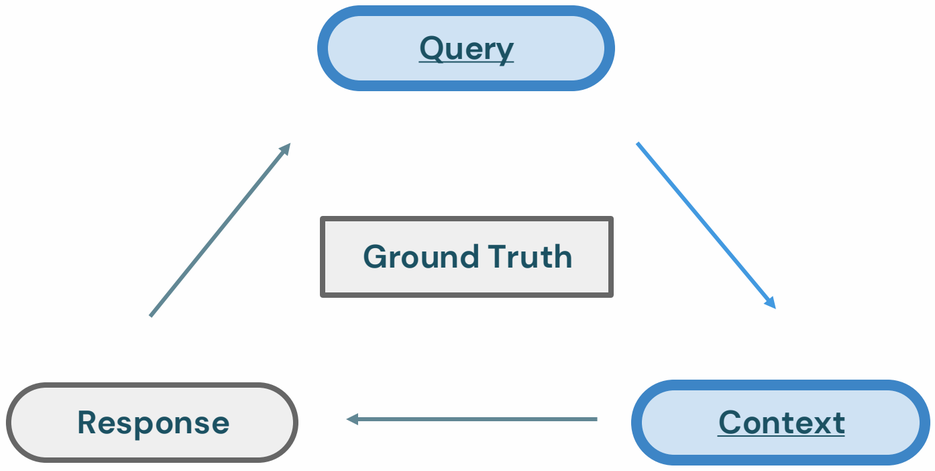
Context Relevance

Retrieval related metrics

Context Relevance:

- Measures **how relevant** the retrieved context is to the query.
- It is evaluated based on the **relationship between** the **query** and the **retrieved context(s)**.
- This metric does not assess factual accuracy but focuses on how well the context supports answering the given question.



Example:

- **Query:** What was Einstein's role in the development of quantum mechanics?
- **High context relevance:** Einstein initially challenged the quantum theory but later contributed foundational ideas to quantum mechanics.
- **Low context relevance:** Einstein was known for his pacifist views during the early 20th century and became a U.S. citizen in 1940.

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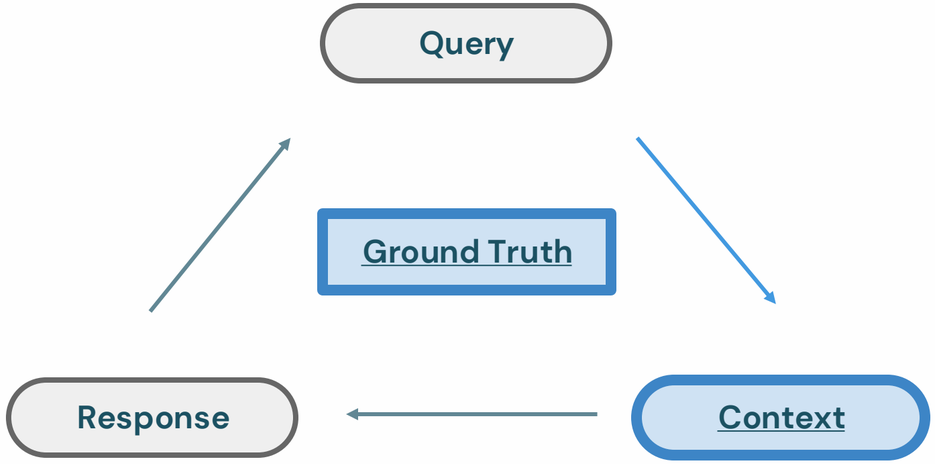
Context Recall

Retrieval related metrics

Context Recall:

- Measures the extent to which all relevant entities and information are retrieved and mentioned in the context provided.
- Based on **Ground Truth** and retrieved **Context(s)**.



Example:

- **Query:** What significant scientific theories did Einstein contribute to?
- **Ground truth:** Einstein contributed to the theories of relativity and had insights into quantum mechanics.
- **High-recall context:** Einstein contributed to relativity and quantum mechanics.
- **Low-recall context:** Einstein contributed to relativity.

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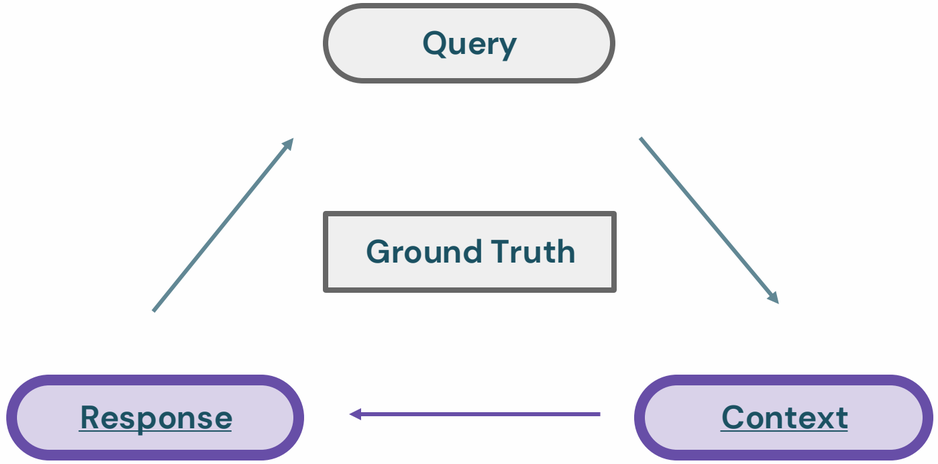
Faithfulness

Generation related metrics

Faithfulness:

- Measures the factual accuracy of the generated answer in relation to the provided context.
- Based on the **Response** and **retrieved Context(s)**.



Example:

- **Query:** Where and when was Einstein born?
- **Context:** Albert Einstein (born 14 March 1879) was a German-born theoretical physicist, widely held to be one of the greatest and most influential scientists of all time.
- **High faithfulness answer:** Einstein was born in *Germany on 14th March 1879*.
- **Low faithfulness answer:** Einstein was born in *Germany on 20th March 1879*.

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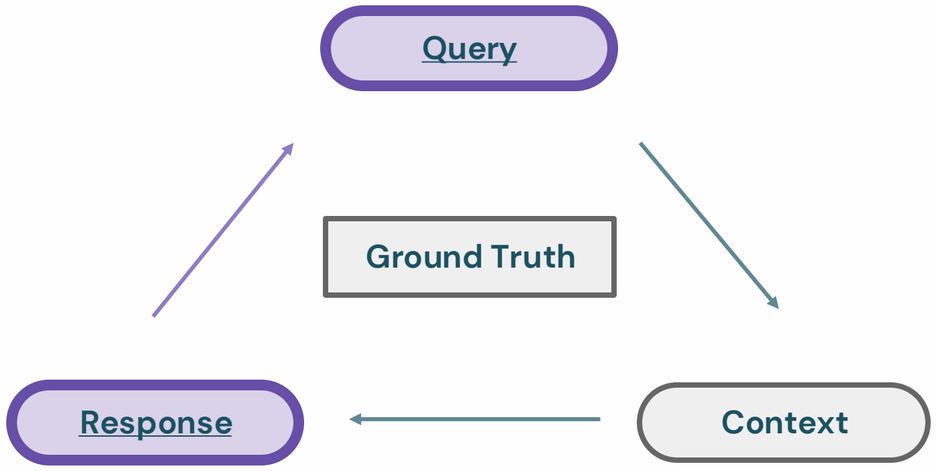
Answer Relevancy

Generation related metrics

Answer Relevancy:

- Assesses how pertinent and applicable the generated response is to the user's initial query.
- Based on the alignment of the **Response** with the user's intent or **Query** specifics.



Example:

- **Query:** What is Einstein known for?
- **High relevancy answer:** Einstein is known for developing the theory of relativity.
- **Low relevancy answer:** Einstein was a scientist.

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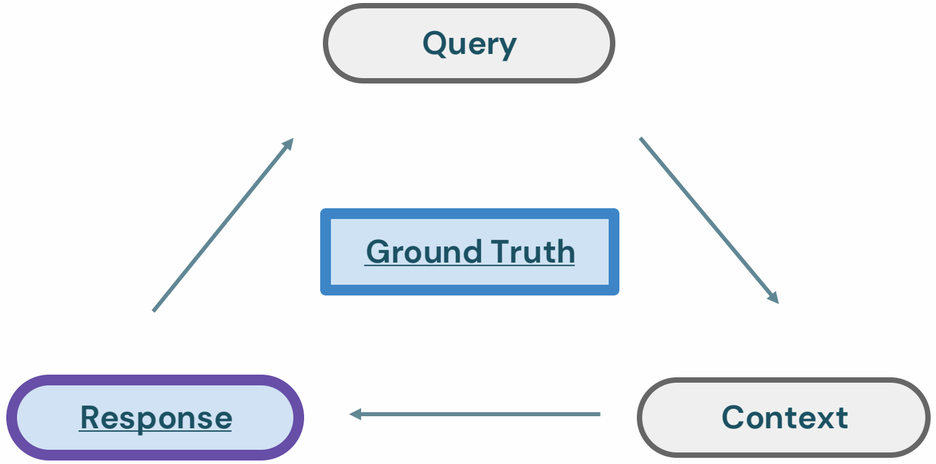
Answer Correctness

Generation related metrics

Answer Correctness:

- Measures the accuracy of the generated answer when compared to the ground truth.
- Based on the **Ground Truth** and the **Response**.
- Encompasses both semantic and factual similarity with the ground truth.



Example:

- **Ground truth:** Albert Einstein was awarded the Nobel Prize in Physics in 1921 for his explanation of the photoelectric effect.
- **High answer correctness:** Einstein received the Nobel Prize in Physics in 1921 for his work on the photoelectric effect.
- **Low answer correctness:** Einstein won the Nobel Prize in Physics in the 1930s for his theory of relativity.

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Custom Metrics

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Custom Metrics for System Evaluation

Frequently related to business goals and constraints for the AI system

- What is important to your use case?
  - Do you care about *serving latency*?
  - Are you concerned with *total cost*?
  - Are you expecting your system to *increase product demand*?
  - What about *customer satisfaction*?
- **Note:** custom metrics can be useful for individual components, too.

Quick Activity: Define your own system-wide custom metric to ensure your AI system is providing the value that you expect.

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Custom Metrics in MLflow

Built-in capabilities for integrating custom metrics into **component monitoring**

- **Custom Guidelines**: Define evaluation rules in natural language for flexible scoring.
- **Unified Scoring**: Combine custom and built-in evaluators in one run.
- **Tracked Results**: Auto-log scores and aggregates in MLflow experiments.

```
from mlflow.metrics.genai import evaluate, Guidelines, RelevanceToQuery, Safety
import mlflow

# (1) Define the professionalism rubric as a Guidelines scorer
professionalism_guidelines = Guidelines(
    name="professionalism",
    guidelines="""Assess the tone of the model's answer for professionalism.

Score using this scale (integer only):
- 0: Extremely casual/informal; slang or colloquialisms; not suitable for professional
contexts.
- 1: Casual but respectful; avoids strong informality; acceptable only in informal
professional settings.
- 2: Mostly formal but with some casual words/phrases; borderline for professional
contexts.
- 3: Balanced, clear, and neutral; suitable for most professional contexts.

Instruction:
- Judge ONLY tone/style, not factual accuracy or relevance.
- Provide a brief justification.
- Return a numeric score (0-3). For pass/fail aggregation, treat scores ≥2 as PASS.
"""
)

# (2) Run evaluation (eval_set has "inputs" and optional "expectations")

with mlflow.start_run(run_name="eval_professionalism") as eval_run:
    eval_results = evaluate(
        data=DA.eval_set,
        predict_fn=predict_fn,
        scorers=[
            professionalism_guidelines,  # replaces the old custom metric
            RelevanceToQuery(),          # optional, built-in
            Safety(),                    # optional, built-in
        ],
    )
```

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Human Feedback and Monitoring

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Human Feedback

Collect data from users and experts

- Often developers are not the experts of the domain
- Models' output need to be **evaluated by human** experts
- Models' outputs and associated feedback need to be collected and stored in a structured manner
- Feedback can be **explicit or implicit**:
  - **Explicit feedback**: Direct and intentional input from users. Such as ratings, comments and review.
  - **Implicit feedback**: Gathered indirectly by observing user behavior and interactions. Such as engagement metrics and behavioral data.

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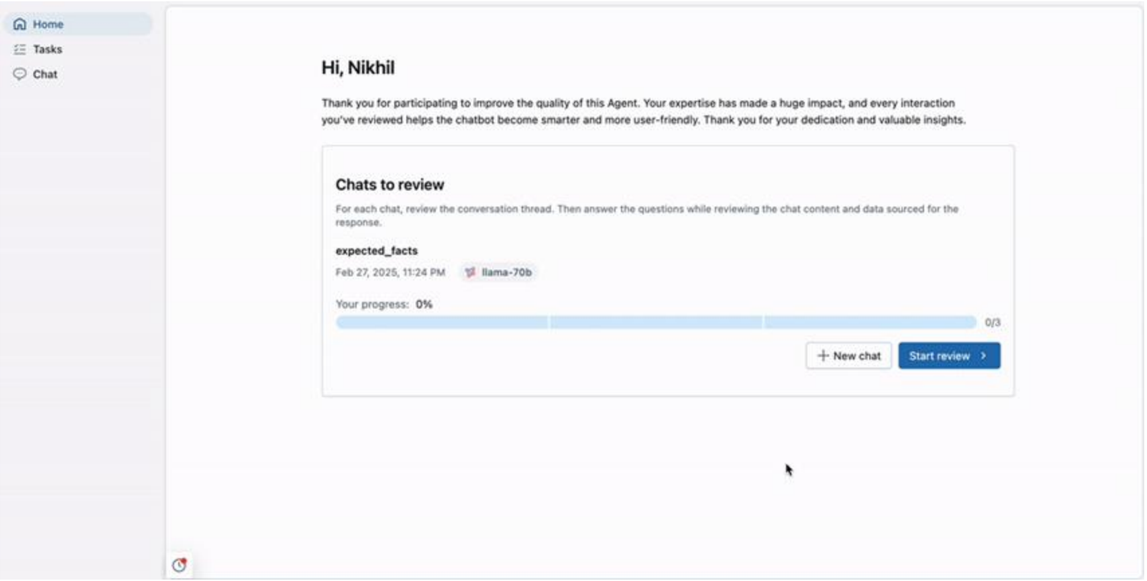
Collect Human Feedback with Review App

Agent Framework provides pre-built feedback app

- Human feedback is essential for building high-quality GenAI applications that meet user expectations.
- Agent Framework allows stakeholders to interact with the agent and **provide feedback**.



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Monitoring Quality in Production

Automatically run quality checks for AI systems in production

- Automatically run MLflow scorers on traces from your production GenAI apps to **continuously monitor quality**.
- Use the **same scorers in development and production** to ensure consistent evaluation.
- Continuous quality assessment with monitoring **running in the background**.

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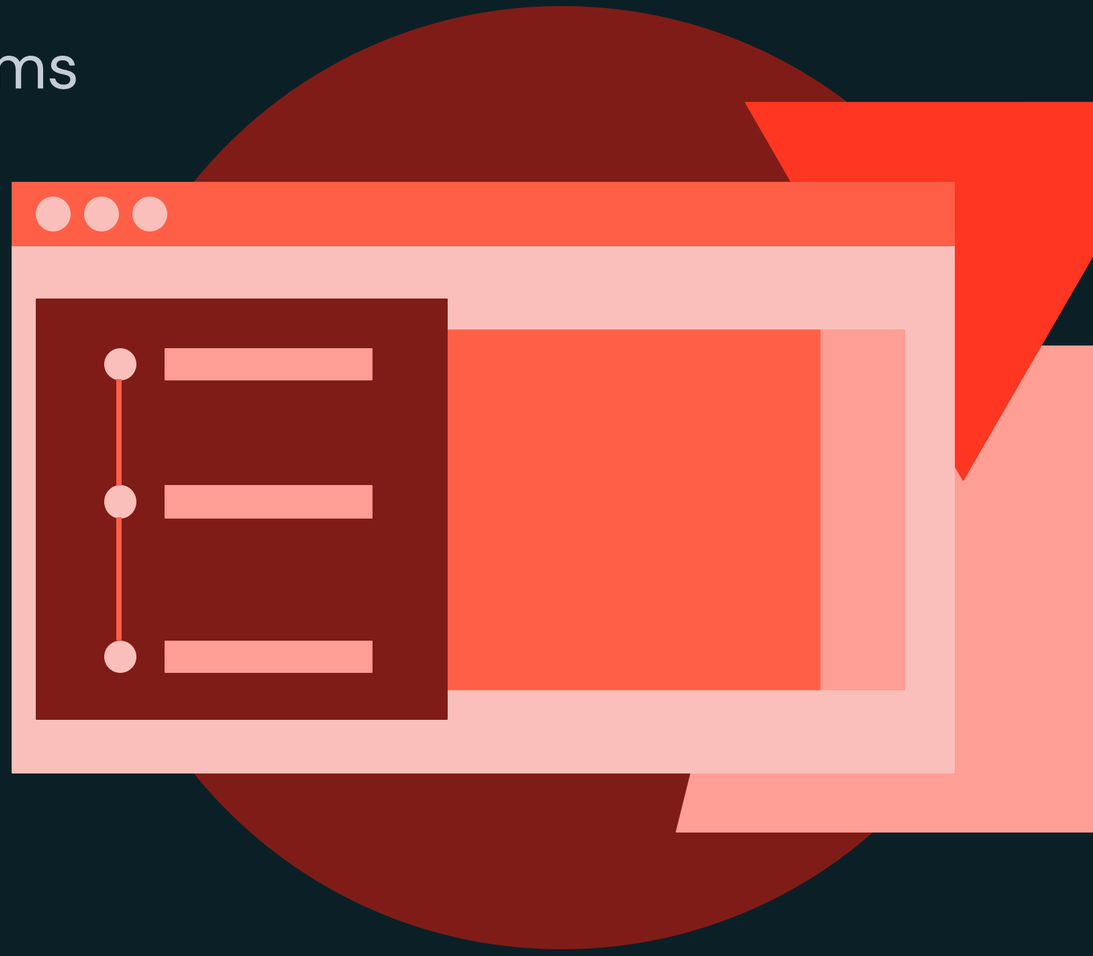
Module 4: Evaluating and Deploying AI Systems

DEMONSTRATION

Evaluation with MLflow



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Demo Outline

Evaluation with MLflow 3.0

- Prepare for evaluation
  - Load the model
  - Load the evaluation dataset
- Model Evaluation
  - Build-in metrics
  - Custom metrics
  - Run Evaluation test
- Review the evaluation results

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Module 4: Evaluating and Deploying AI Systems

LECTURE

Real-time Deployment with Model Serving

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Deployment Paradigms*

Similar to traditional machine learning extended to GenAI

Latency Decreases

Throughput Decreases**

Batch	Streaming	Real-time
Generate (and store) completions on a table of (text) inputs/prompts	Generate (and store) completions on micro-batches of (text) inputs/prompts as they're processed	Generate completions asynchronously in real-time on individual input/prompt

* **Edge Deployment:** Edge (on-device) deployment is challenging with large language models due to space requirements.

** You can have high-throughput real-time serving by scaling out across a bunch of machines ("concurrency").

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Challenges with Deploying Real-time AI

Most AI systems don't get into production

Infrastructure is hard

Real-time AI systems require fast and scalable serving infrastructure, which is costly to build and maintain

Deploying real time models needs disparate tools

Data teams use diverse tools to develop models

Customers use separate platforms for data, LLMs and Serving, adding complexity and cost

Operating production AI requires expert resources

Steep learning curve of deployment tools.

Model deployment is bottlenecked by limited engineering resources, limiting the ability to scale AI

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Mosaic AI Model Serving

Integrate your model into your websites and applications as an API

Production-Grade Serving

Highly available, low latency, scalable serving that works for small and large workloads

Accelerate deployments with Lakehouse-Unified Serving

Automatic feature lookups, monitoring and unified governance that **automates deployment** and reduce errors

Simplified Deployment

Simple and flexible deployment through **UI** or **API**

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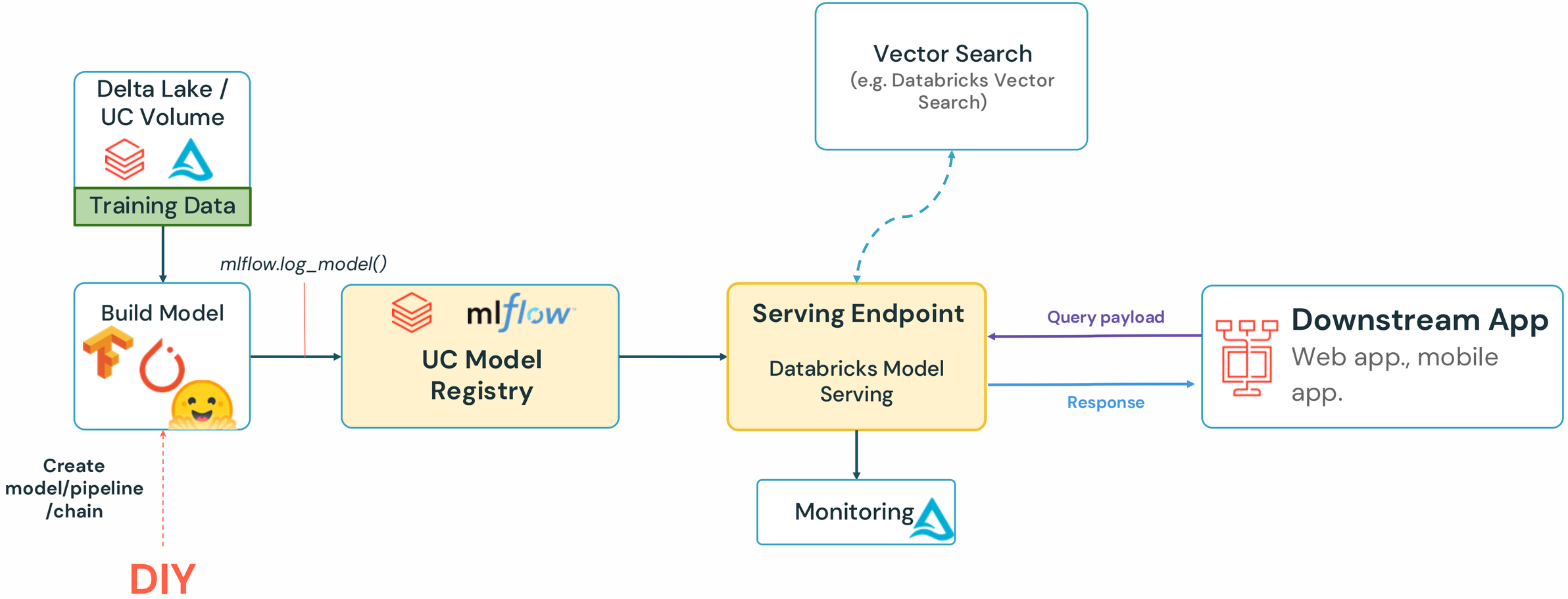
Real-time Deployment

A typical real-time model deployment workflow for AI systems



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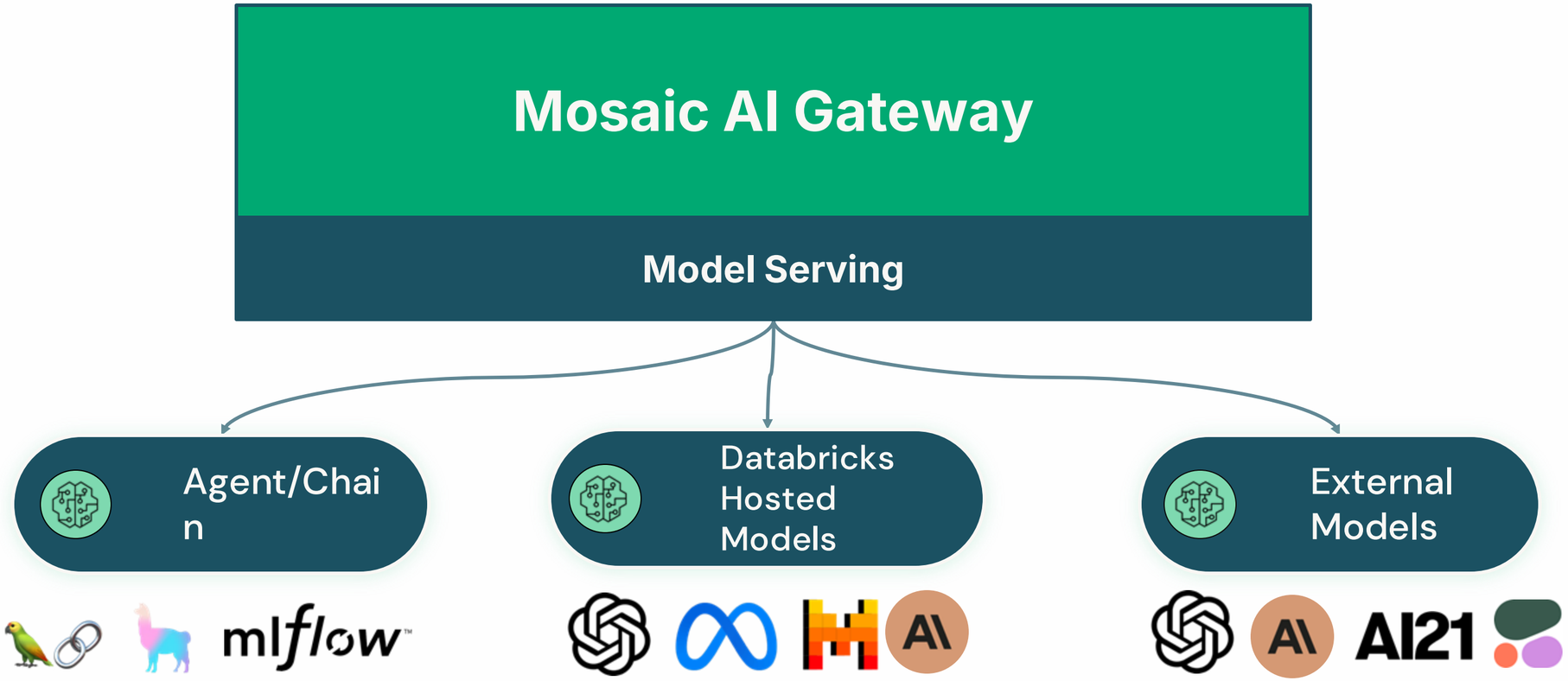
Mosaic AI Gateway

Protect data and Gen AI deployments through centralized governance



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Features

- **Usage tracking and observability**: Monitor who's using models and how often to help optimize performance and manage costs.
- **Payload logging**: Capture requests and responses in Unity Catalog, allowing you to debug and improve model quality.
- **Unified Guardrails**: A single source of truth for data policies, with guardrails like PII detection to ensure compliance.
- **A/B testing and traffic policies:** Fallback/retry policies.
- **Permissions and granular rate limiting**: Only authorized users can access models and APIs, while controlling the number of requests to optimize performance.

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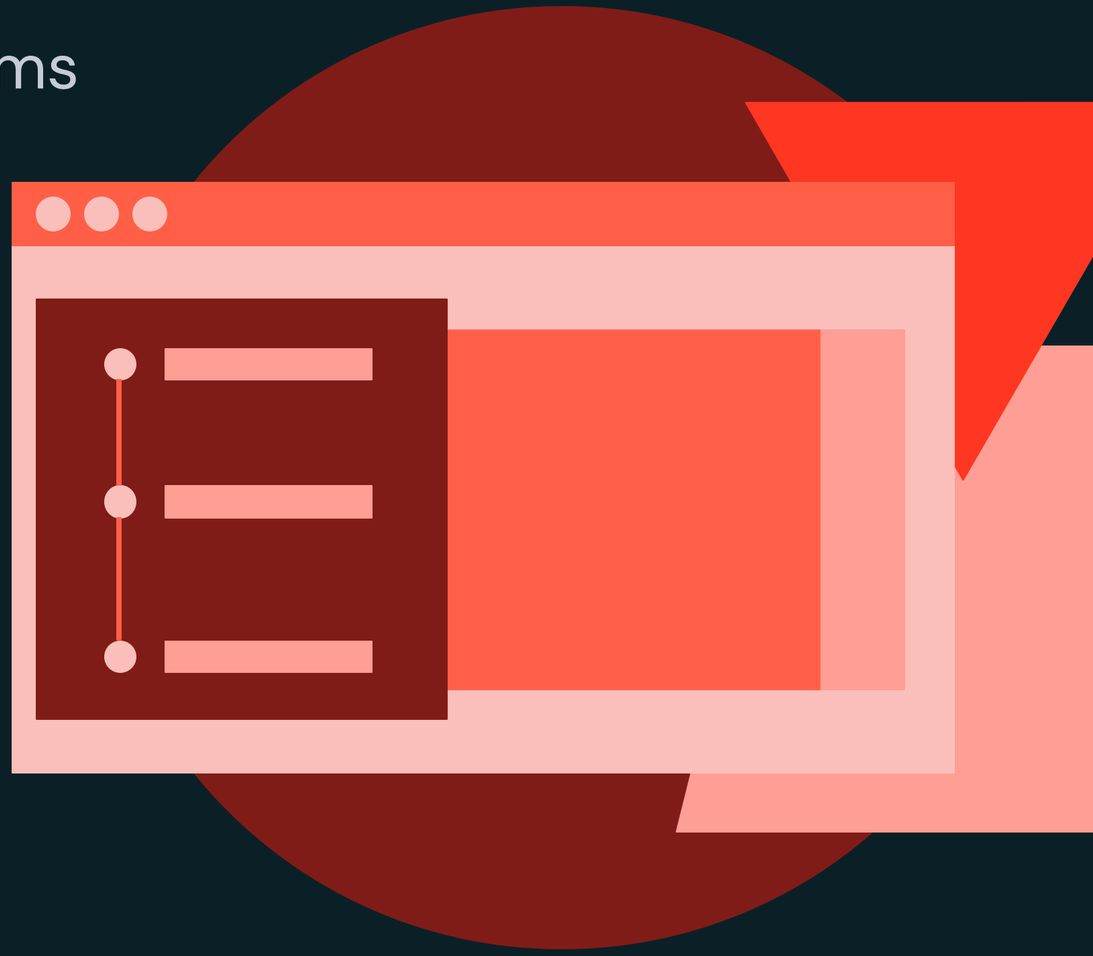
Module 4: Evaluating and Deploying AI Systems

DEMONSTRATION

Real-time Deployment with Model serving



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Demo Outline

External Models with AI Gateway

- Serve the model with Agent Framework
- Test the model with Review App
- Collect human feedback via Review App

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Agent Bricks

Get Started with Databricks for Generative AI

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Objectives

- Discuss the challenges of building GenAI systems.
- Explain how Agent Bricks addresses these challenges.
- Define core features of Agent Bricks.
- Distinguish the key Agent Bricks task types and their common use cases.

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Module 5: Agent Bricks

LECTURE

Introduction to Agent Bricks

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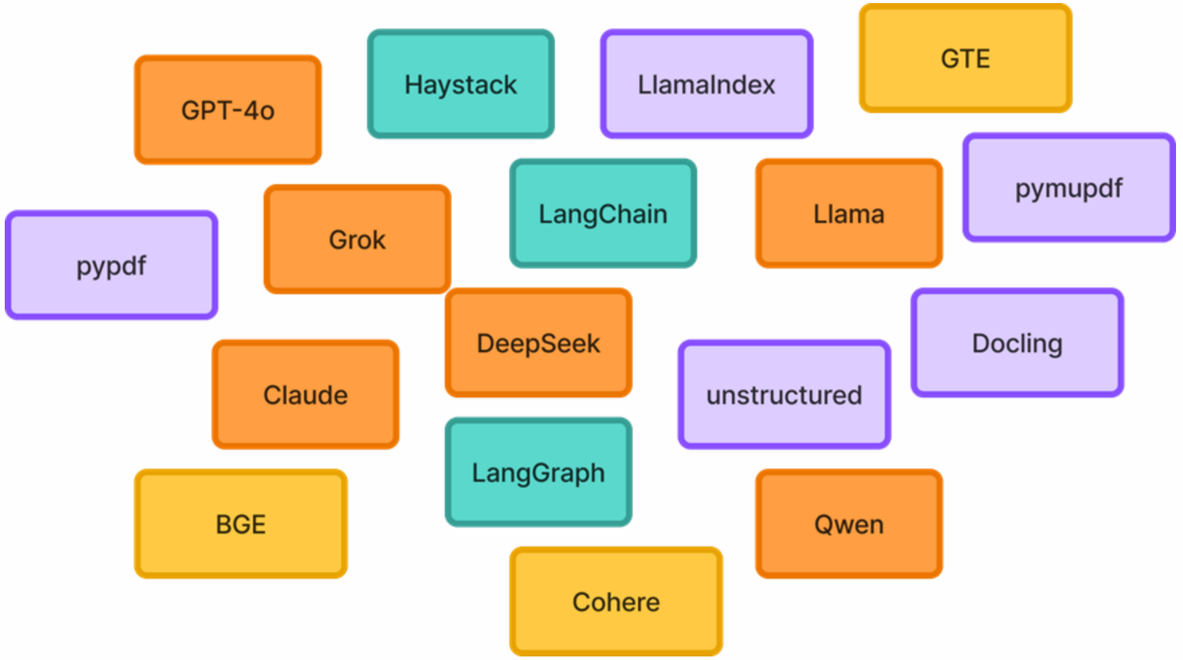
Challenges for Building GenAI Systems

- **How do you adapt AI systems to your needs?**
  - LLMs do not understand enterprise data well enough
  - Prompt engineering is often based on “vibe checks”
  - Too many tools/components to research and integrate
- **How do you get to the right quality?**
  - Lack of labeled data
  - Missing verifiable custom evaluations
- **Does the ROI make sense?**
  - Larger models are expensive
  - Will end users keep using the product?
  - Does the output quality justify the cost?



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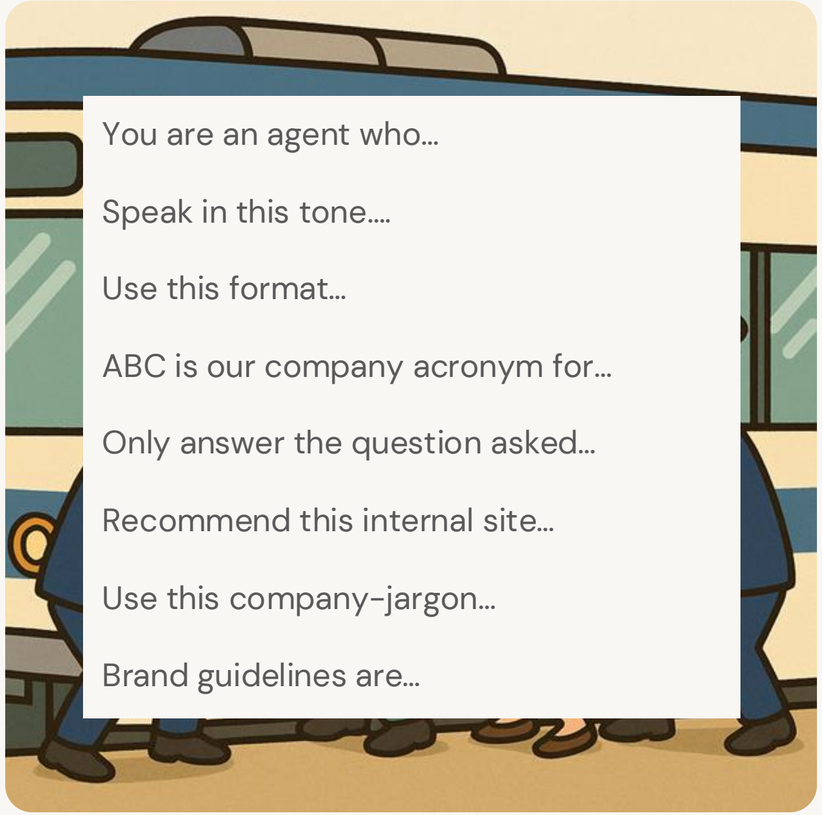
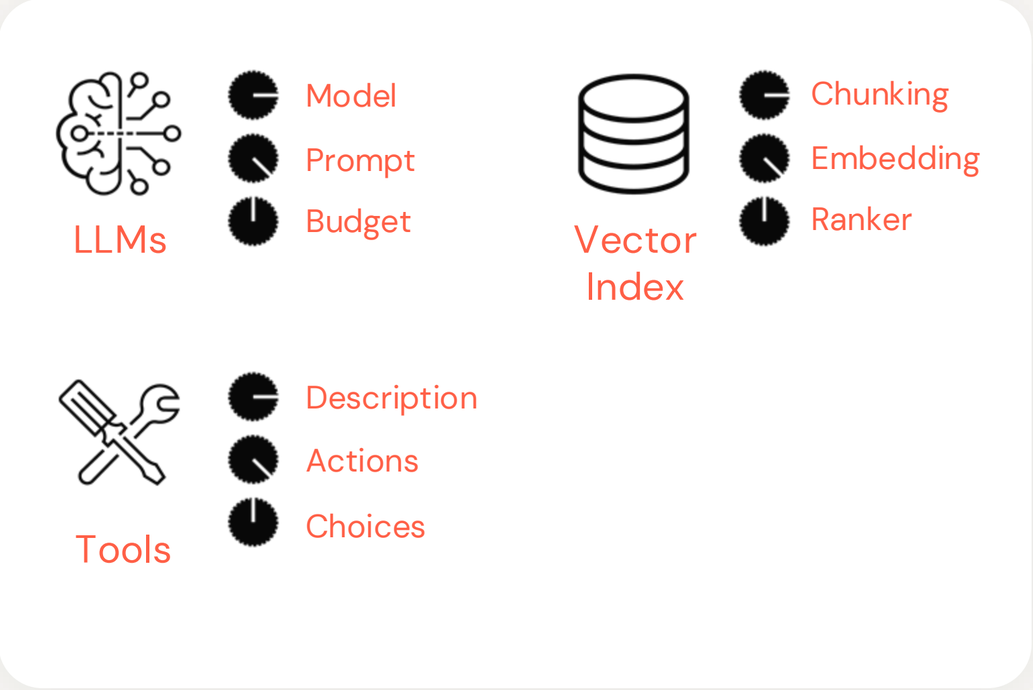
Example: RAG Pipeline

"LLM Prompt Stuffing"



But an a RAG pipeline has many components and knobs...



👍/👎 Evaluation is manual and optimizing LLM based on feedback is complex

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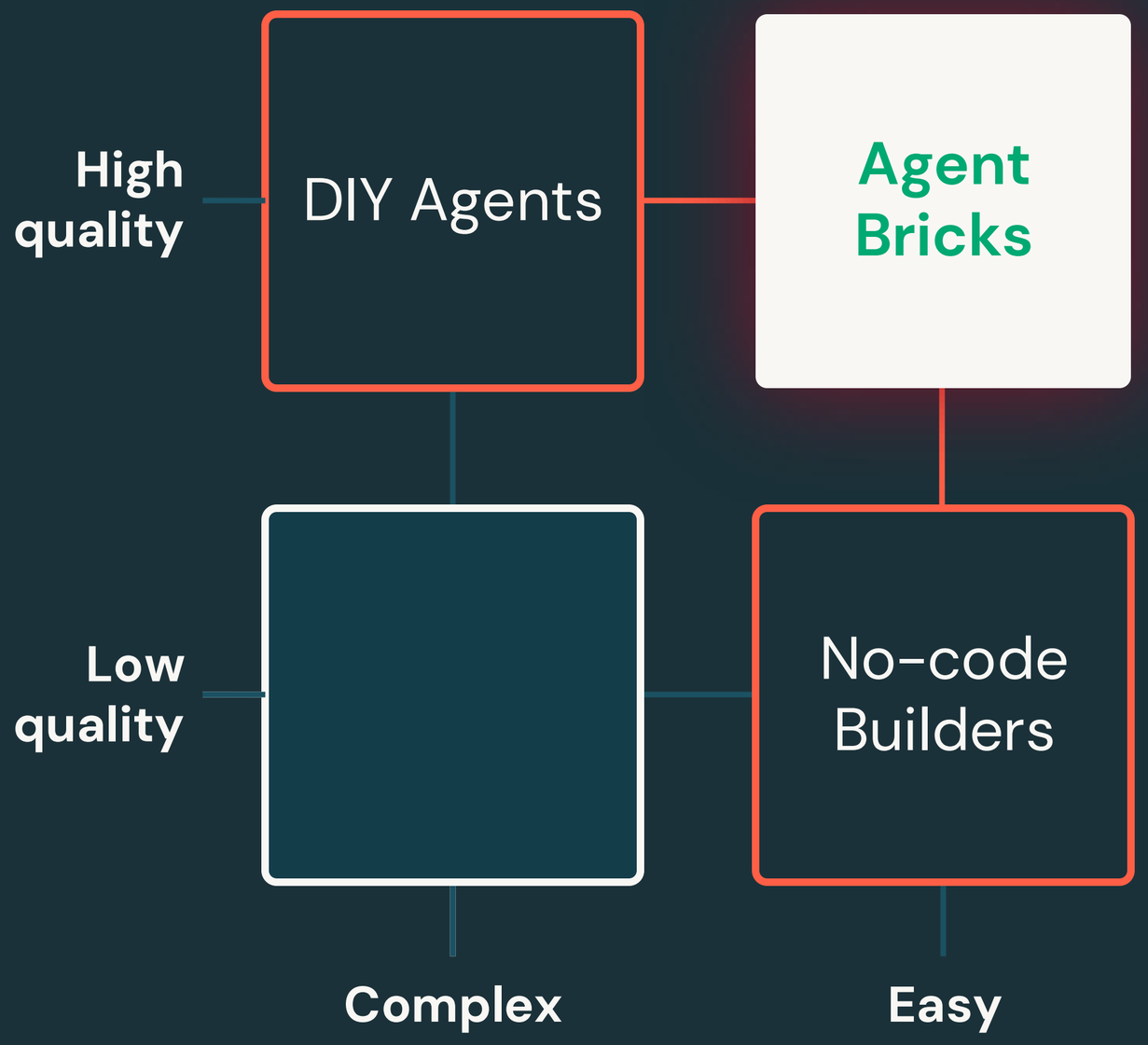

High quality
Low quality
DIY Agents
Agent Bricks
No-code Builders
Complex
Easy

Best-of-N

Adaptive LLM judges

Test-time adaptive optimization

Constitutional AI

Deep research agents

MixAttention

Fine-tuning

Value-guided decoding

Critique filtering

RLHF

Prompt optimization

Synthetic data generation

Agent Bricks

Automated evaluation

Powered by Databricks Mosaic AI Research

Multi-agentic patterns

Agentic knowledge graphs

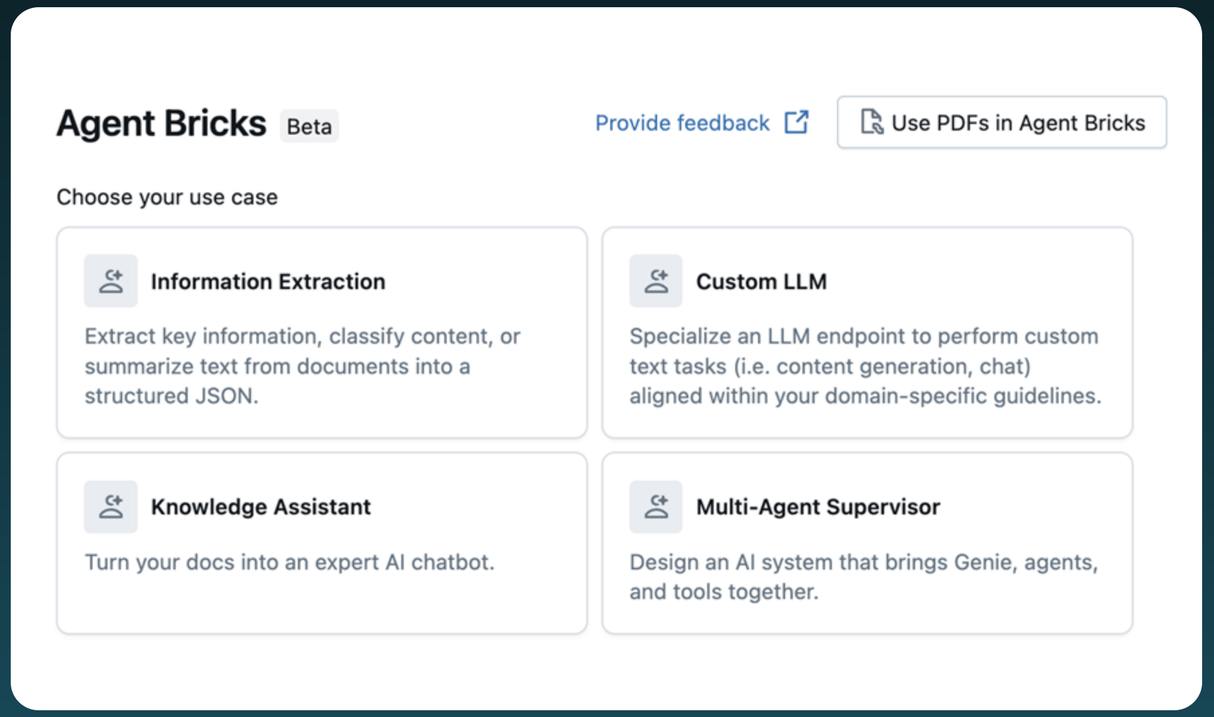
o1-style reasoning

Contextual retrieval

RLVR

Agent Bricks

1 Select a task, declare a high-level description of the agent



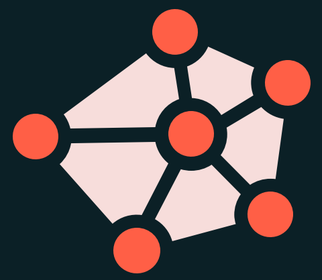
Agent Bricks

1. Select a task, declare a high-level description of the agent
2. Agent Bricks auto-magically:
   - Creates evaluation benchmarks

Automatically create quality benchmarks on **your task**

- Accuracy
- Product Relevance
- Customer Churn

Generate new training data



Agent Bricks

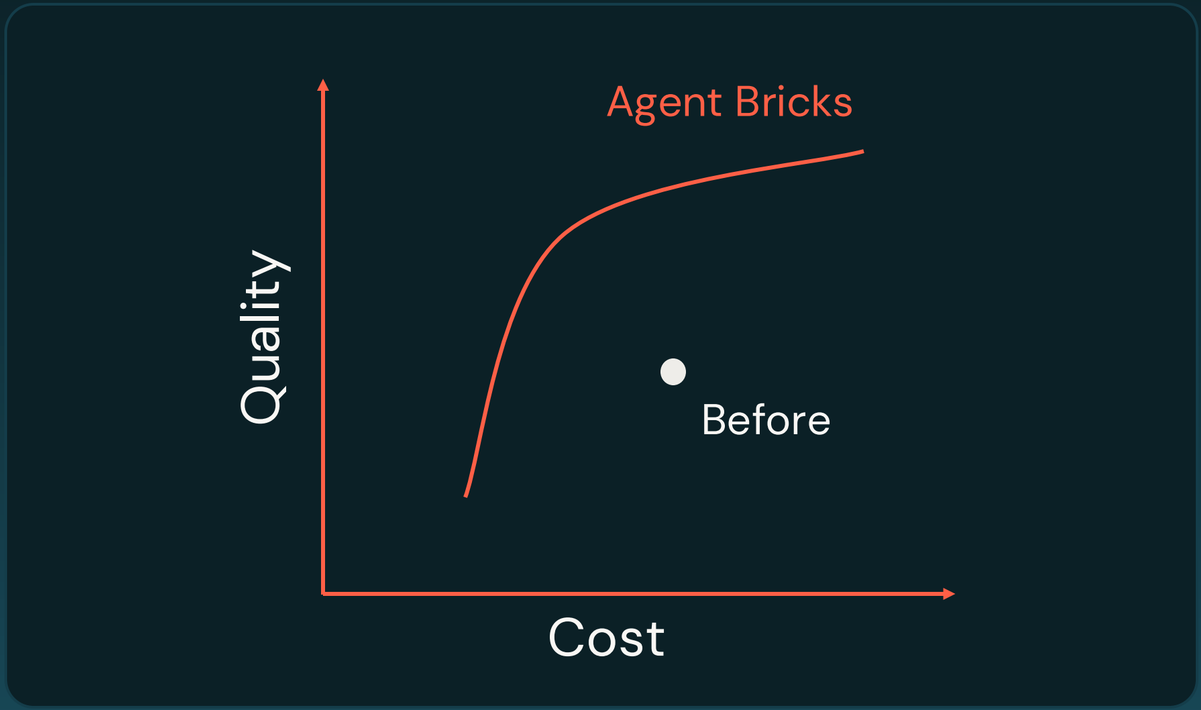
1. Select a task, declare a high-level description of the agent
2. Agent Bricks auto-magically:
   - Creates evaluation benchmarks
   - Auto-optimizes the agent

Searches and compose different techniques for quality

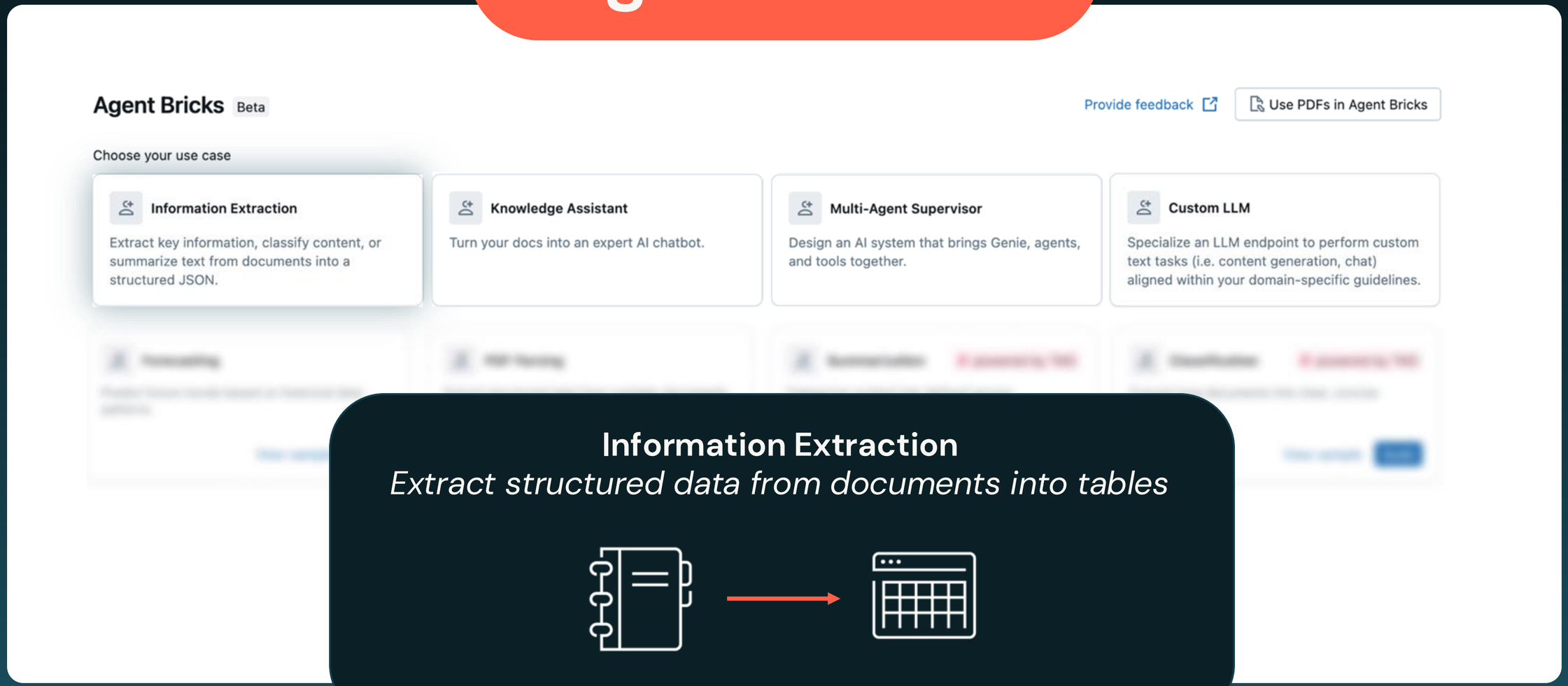
- Prompt optimization
- Fine-tuning
- Tool choices
- Reward model filtering
- Custom LLM Judges
- RLHF

Agent Bricks

1. Select a task, declare a high-level description of the agent
2. Agent Bricks auto-magically:
   - Creates evaluation benchmarks
   - Auto-optimizes the agent
   - Cost *and* Quality
3. Deploy and iterate on quality



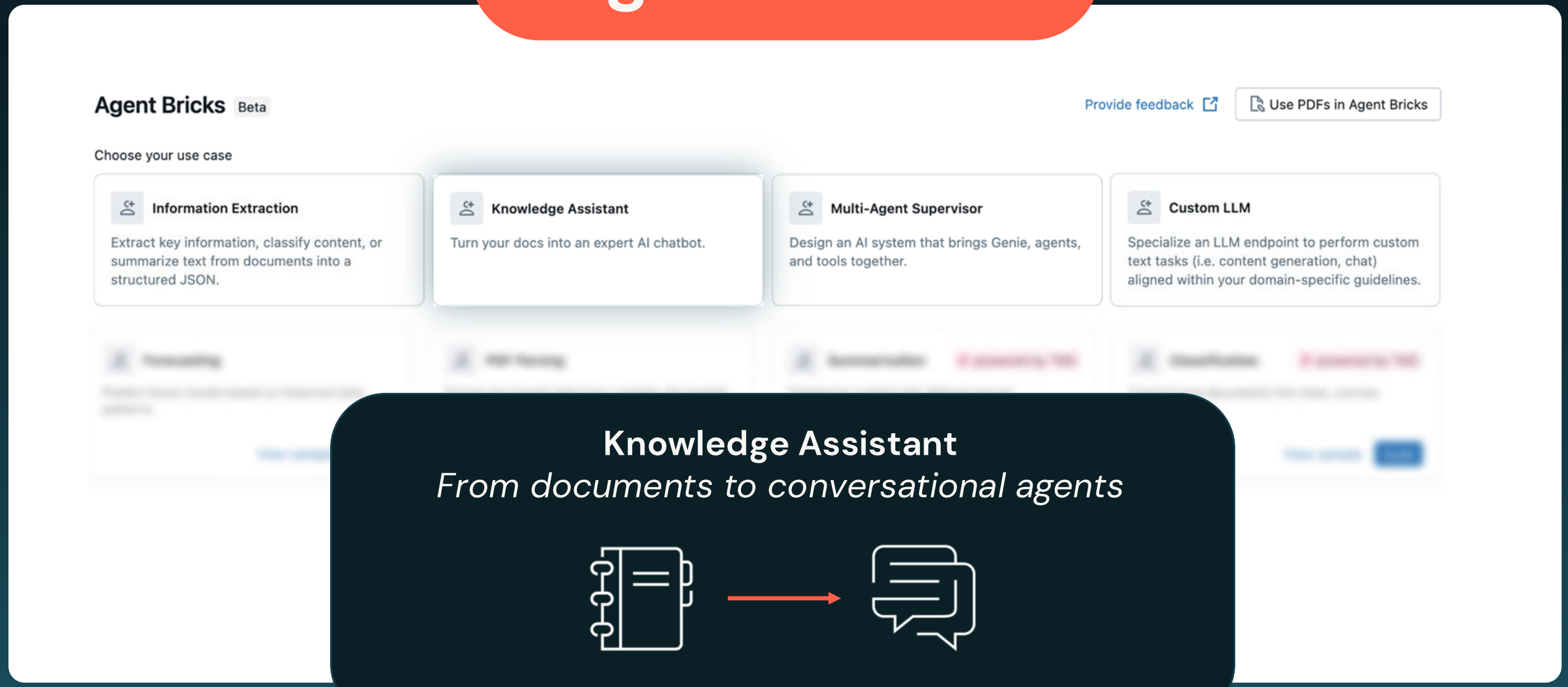
Agent Bricks



Information Extraction

Extract structured data from documents into tables

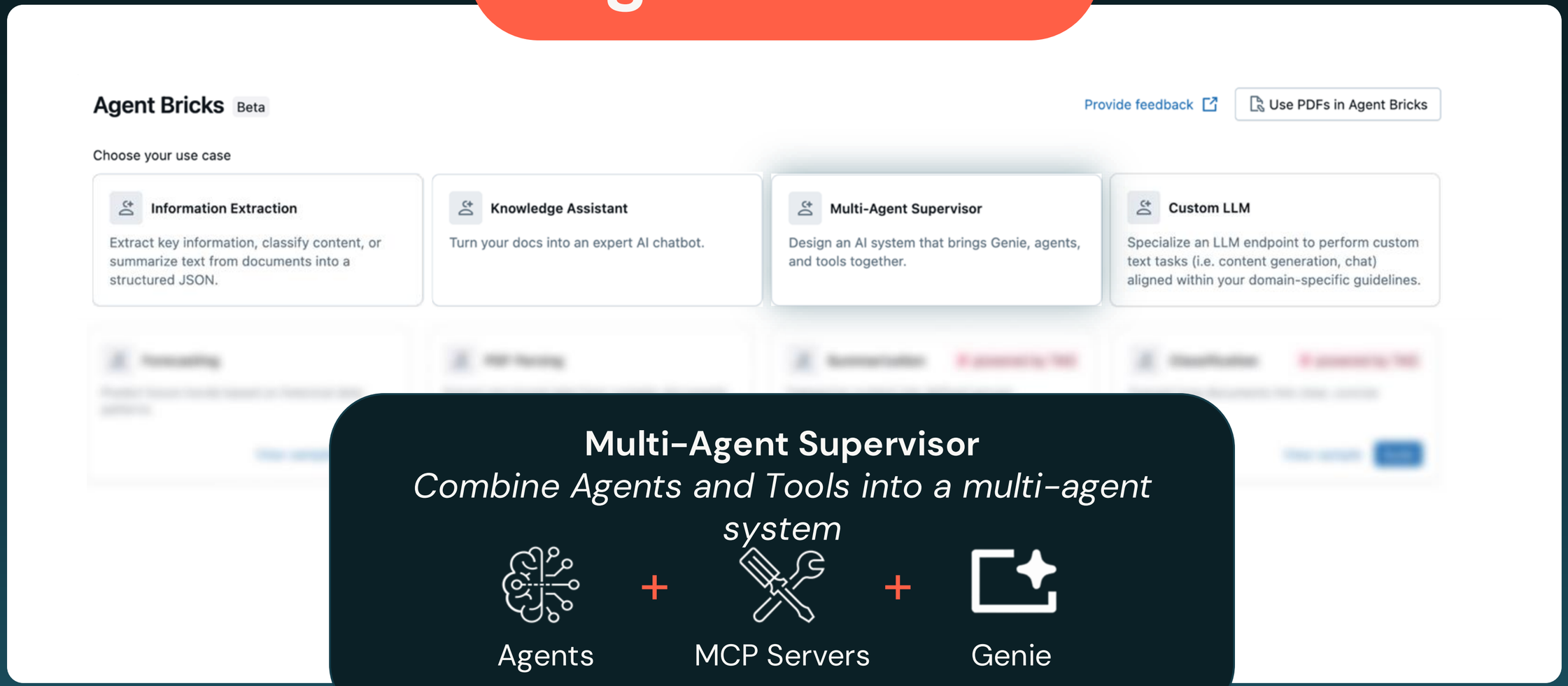
Agent Bricks



Knowledge Assistant

From documents to conversational agents

Agent Bricks

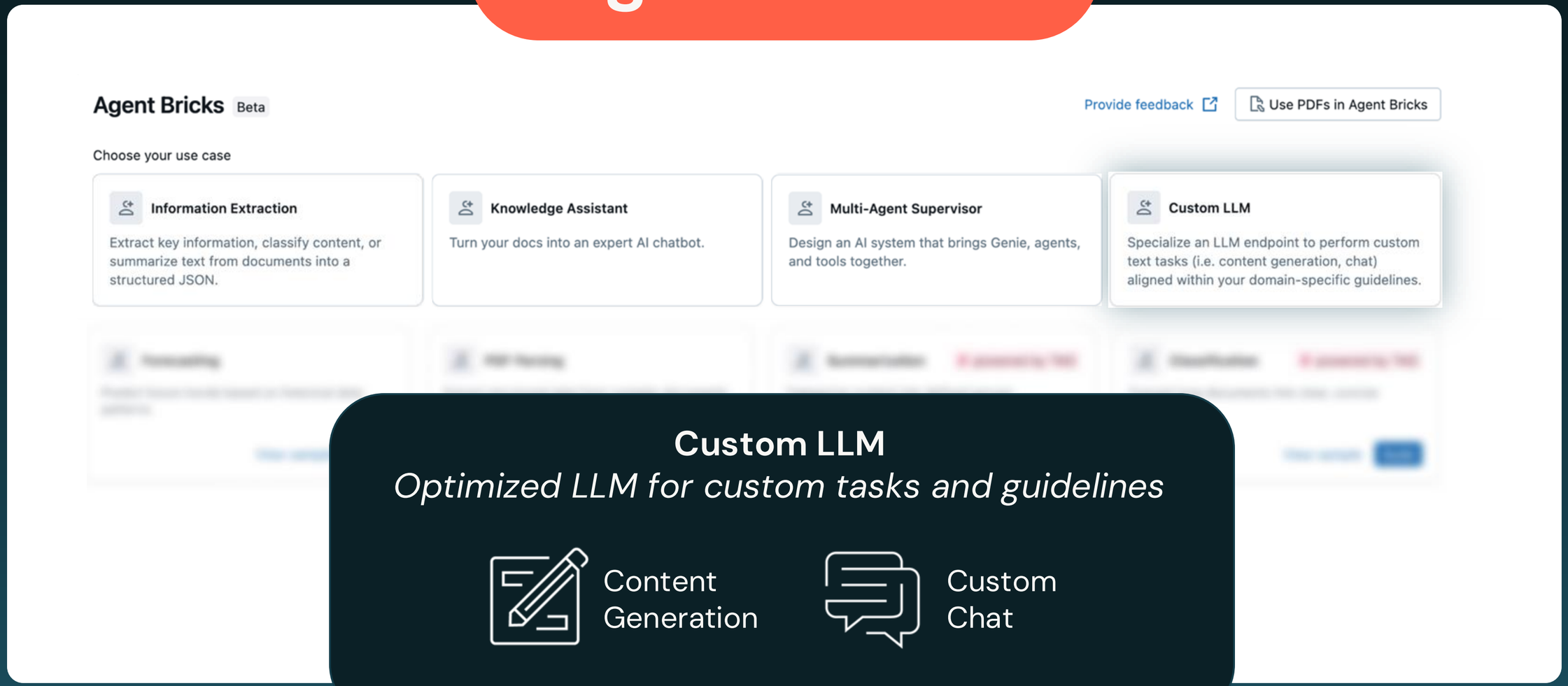


Multi-Agent Supervisor

Combine Agents and Tools into a multi-agent system

Agents + MCP Servers + Genie

Agent Bricks



Custom LLM

Optimized LLM for custom tasks and guidelines

Content Generation

Custom Chat

Use Cases

Agent Bricks

Information Extraction Agent

Problem: Data is buried

Value: Turn unstructured text into structured fields like names, dates, and entities—no labeled data needed.

Custom LLM Agent

Problem: Teams rely on generic models or manual effort to tailor text for their domain.

Value: Transform text for tasks like summarization, classification, or rewriting—optimized for your industry.

Knowledge Assistant Agent

Problem: Chatbots give vague or incorrect answers

Value: Deliver fast, accurate answers grounded in your enterprise data.

Multi-Agent Supervisor

Problem: Complex tasks break down when a single model tries to do everything at once.

Value: multi-agent system with higher-quality results with less manual debugging or retraining

LAB EXERCISE

End-to-end RAG Pipeline

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Lab Outline

In this lab, you will build a Retrieval-Augmented Generation (RAG) pipeline on Databricks using various generative AI tools. The RAG pipeline will leverage Databricks documentation as an external knowledge base to answer questions related to Databricks.

Task Outline:

- Create a vector search index.
- Build the RAG model.
- Register the RAG model.
- Evaluate the RAG model.
- Analyze and interpret the evaluation results.
- Deploy the model.
- Test the model using the Review App.

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Summary and Next Steps

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Next Steps

Additional resources for continuing the learning journey

Generative AI Engineering with Databricks

- Continue your learning through self-paced or instructor-led offerings
- These learning pathways include following courses;
  - Generative AI Solution Development
  - Generative AI Application Development
  - Generative AI Application Evaluation and Governance
  - Generative AI Application Deployment and Monitoring

Generative AI Engineer Associate Certification

- Validate your generative AI skills on Databricks by earning a Databricks credential
- Exam information and exam guide
- The exam covers:
  - Design Applications – 14%
  - Data Preparation – 14%
  - Application Development – 30%
  - Assembling and Deploying Apps – 22%
  - Governance – 8%
  - Evaluation and Monitoring – 12%

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